



ALL LIFE BANK CUSTOMER SEGMENTATION

UNSUPERVISED LEARNING PROJECT



MARCH 2, 2025

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CONTEXT

AllLife Bank is a financial institution that is looking to enhance its focus on its **credit card customer base** in the upcoming financial year. The bank has received insights from its **marketing research team**, which suggest that the **market penetration** for credit card services can be improved. To achieve this, the marketing team proposes running **personalized campaigns** to target both **new customers** and **existing customers**. These campaigns aim to:

- **Acquire new customers:** Attract individuals who do not currently hold a credit card with AllLife Bank.
- **Upsell to existing customers:** Encourage current credit cardholders to use more services or upgrade their existing credit card plans.

Additionally, the research revealed that customers perceive the bank's **support services** as subpar. This negative perception could be a barrier to customer satisfaction and retention. To address this, the **Operations team** plans to **upgrade the service delivery model** to ensure that customer queries are resolved more efficiently and effectively.

Given these challenges, both the **Head of Marketing** and the **Head of Delivery** have decided to collaborate with the **Data Science team** to leverage data-driven insights for improving customer engagement and service delivery.

OBJECTIVE

The primary objective of this project is to **identify different segments** within the existing customer base based on their **spending patterns** and **past interactions** with the bank. By segmenting customers, the bank can:

1. **Understand Customer Behavior:** Gain insights into how different groups of customers interact with the bank, their spending habits, and their preferences.
2. **Personalize Marketing Campaigns:** Tailor marketing strategies to target specific customer segments with relevant offers, promotions, and communication.
3. **Improve Service Delivery:** Identify areas where customer support can be enhanced to address the specific needs of different customer groups.
4. **Enhance Customer Retention:** By understanding the needs and behaviors of different segments, the bank can take proactive steps to retain valuable customers and reduce churn.

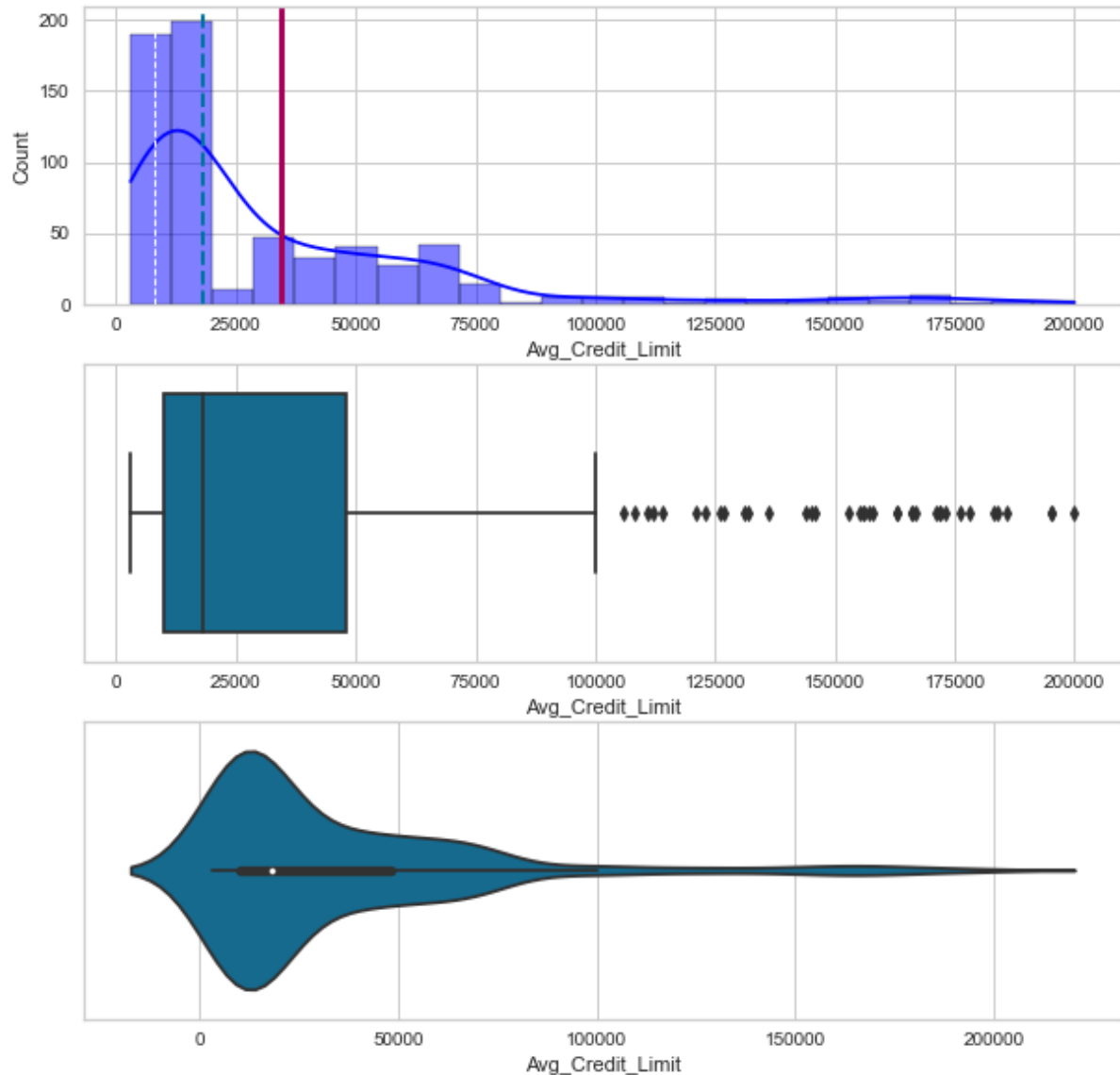
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Univariate analysis

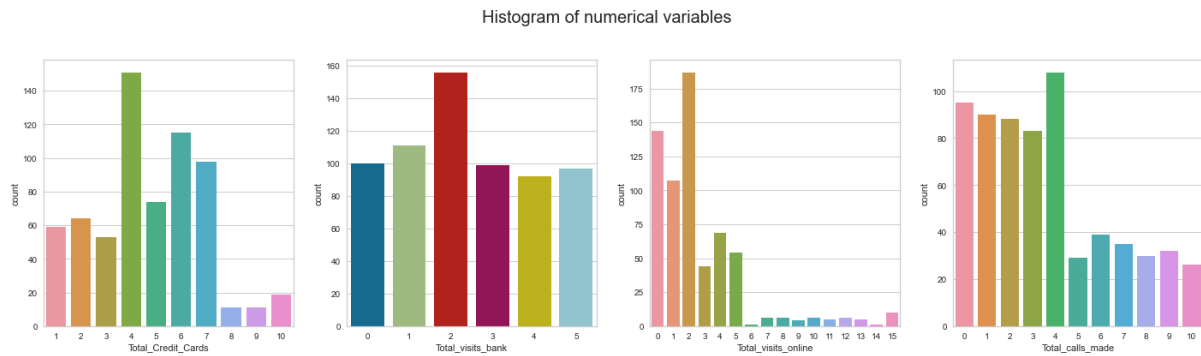
Distribution of the Avg_Credit_Limit



Observations:

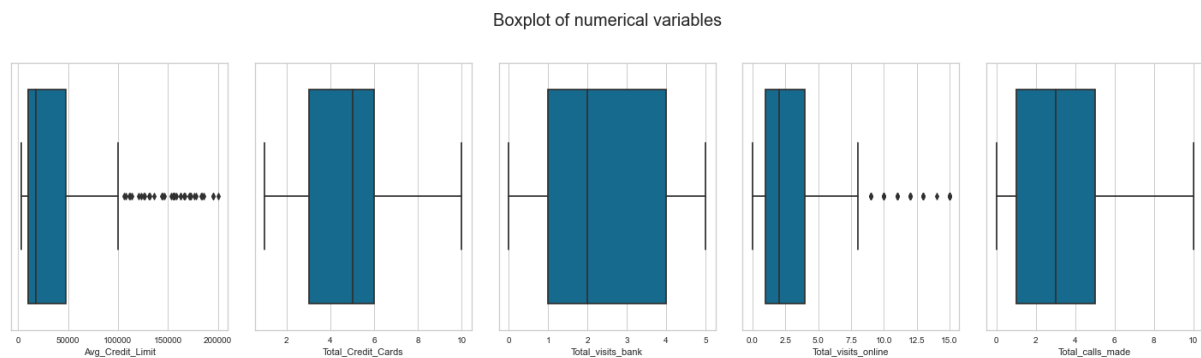
- The Avg_Credit_Limit of a customer varies from 3000 to 200000.
- Avg_Credit_Limit is highly right skewed.
- There are lot of outliers to the right.
- The mean of Avg_Credit_Limit is 34574 while the median is 18000.
- The 3rd quantile value is at 48000 while the first quantile value is at 10000 and the median is at 18000.
- Majority of the customers have Avg_Credit_Limit lesser than the median value-18000.

- There are quite some customers whose average credit limit is higher than 100000.



Observations:

- There are more customers who have 4 to 7 credit cards.
- 67.18% of customers made atleast 2 visits to the bank.
- Most of the customers have made atleast 2 visits online.
- Most of the customers have made 4 calls to the bank.

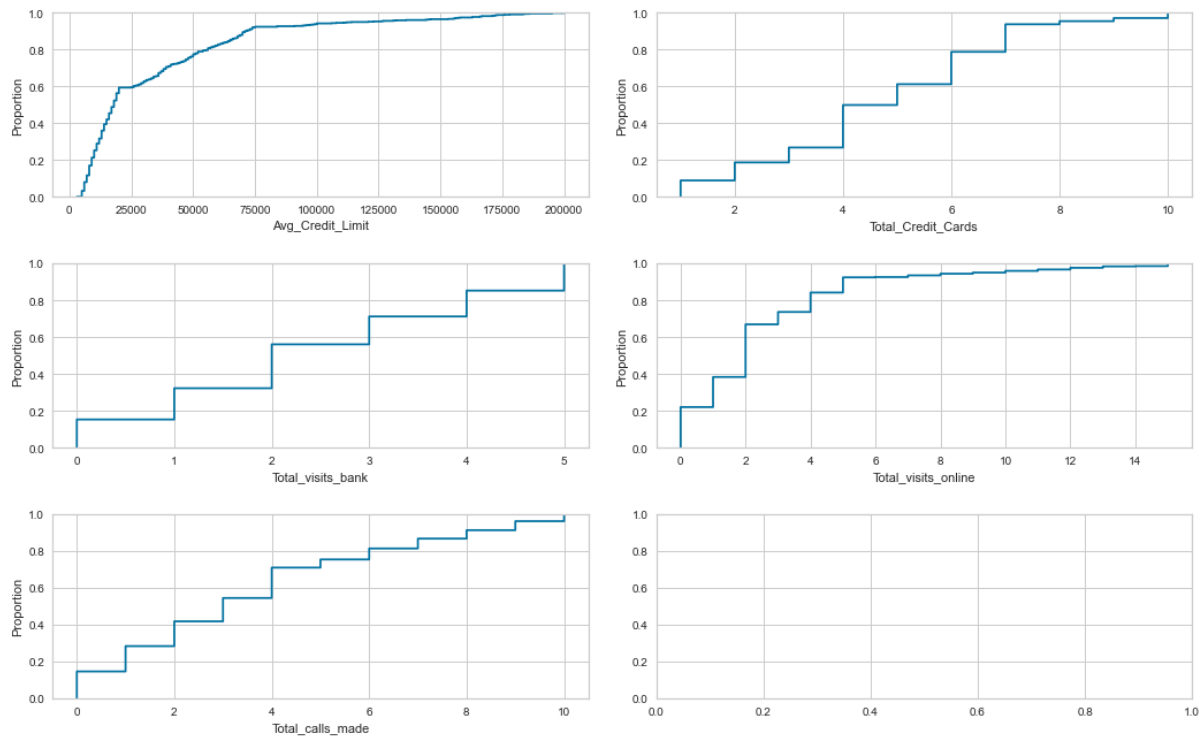


Observations:

- There are upper outliers in the Avg_Credit_Limit which indicates that the credit limit of few customers is very high.
- 50% of the customers have Avg_Credit_Limit lesser than 18000 while the max is 200000.
- 50% of the customers have atleast 3-6 credit cards.
- 50% of the customers have made 2 visits to the bank.

- There are upper outliers in the Total_visits_online which indicates that those customers preferred mode of querying the bank is through online, for various reasons. Probably the customers didn't get the satisfactory response and that is why they are making multiple calls to the bank's customer support.
- 75% of the customers have made up to 4 visits online while the max is 15 visits.
- 75% of the customers have made 5 calls and the max is 10 calls.

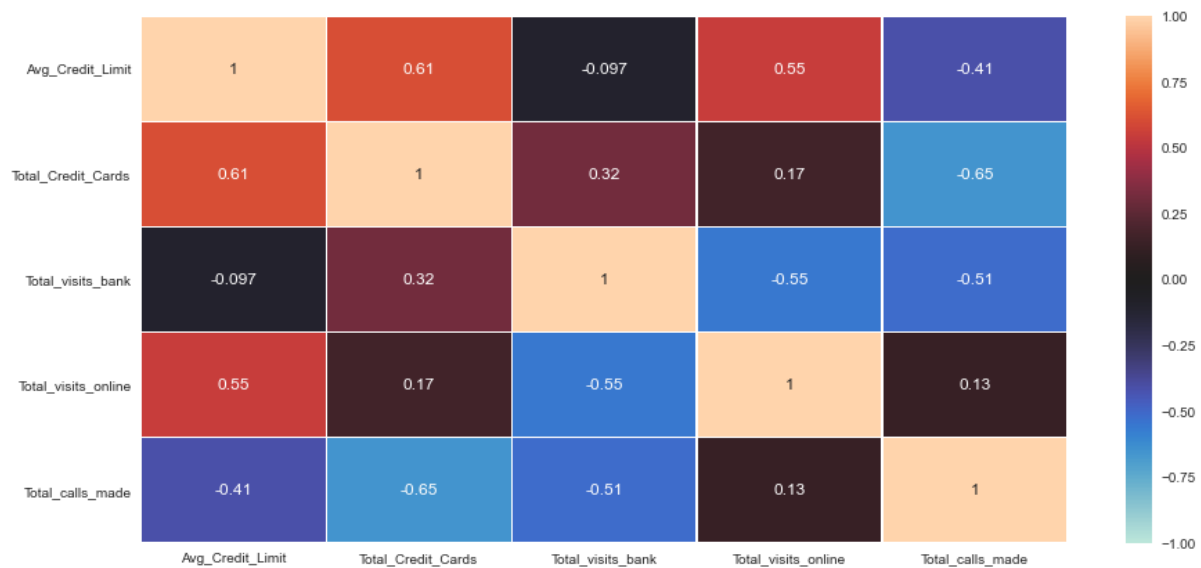
CDF plot of numerical variables



Observations:

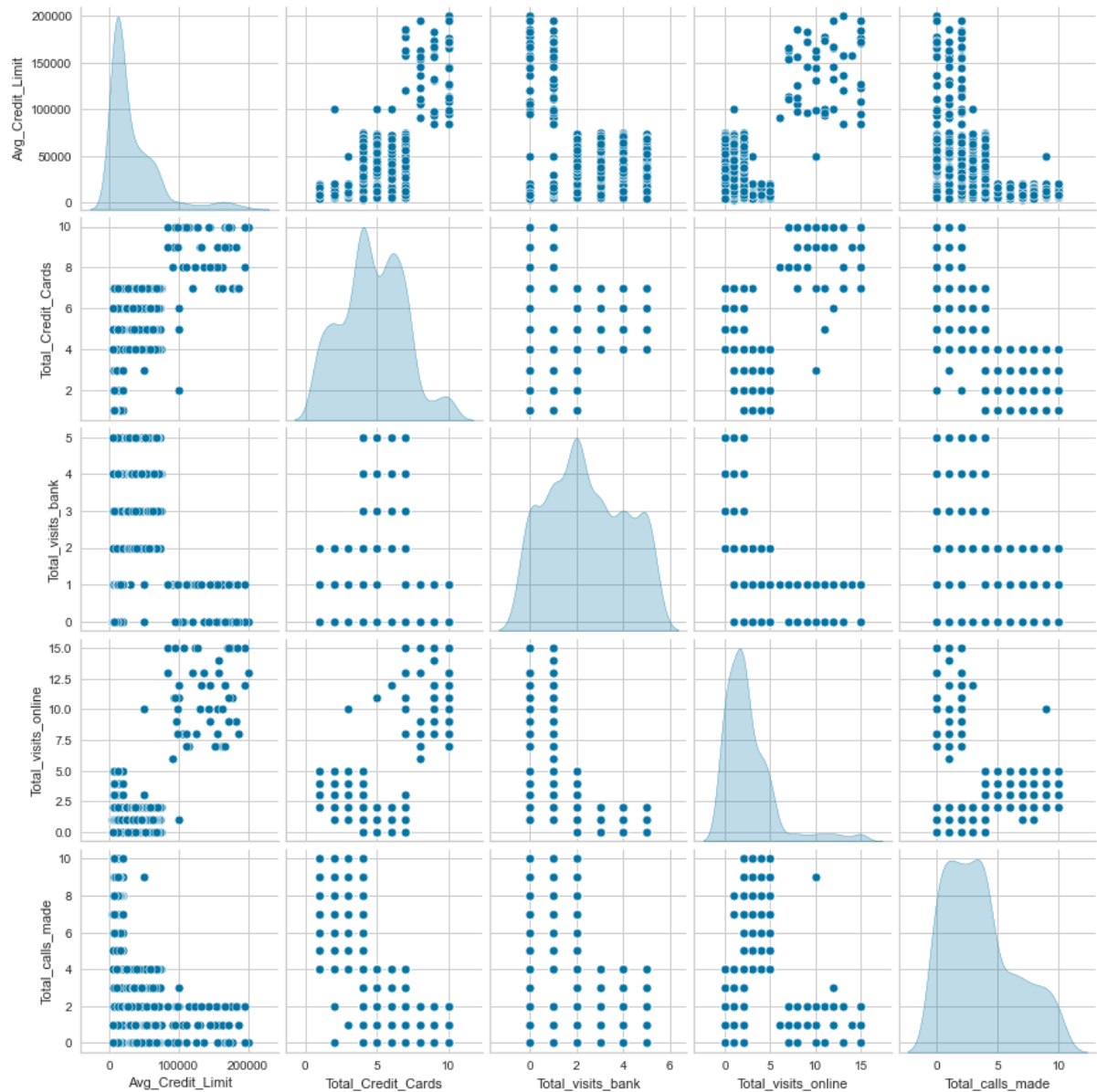
- We notice that around 90% of the customers have average credit limit below 75,000.
- 60% of the customers have at most 6 credit cards.
- 90% of the customers have made 5 visits online.
- 80% of the customers have made 6 calls to the bank for enquiries.
- 50% of the customers have made 2 visits to the bank.

Bivariate Analysis

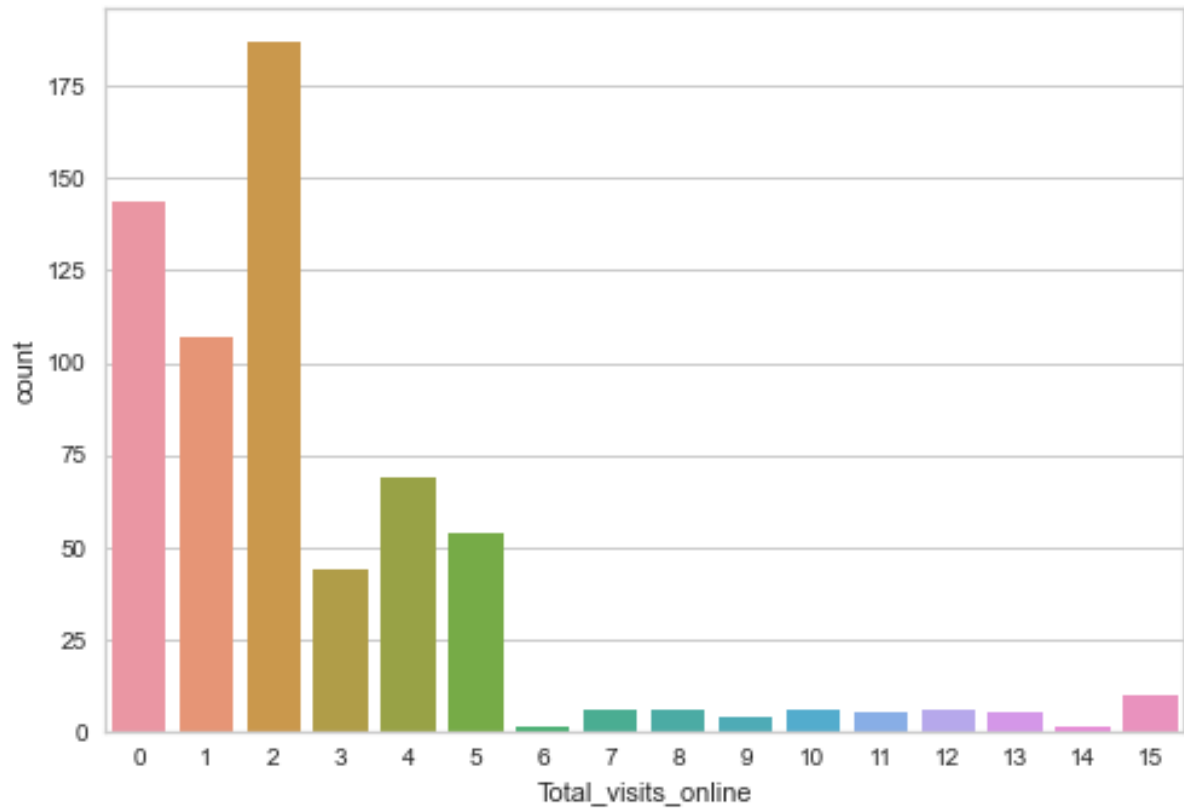


Observations:

- Avg_Credit_Limit has strong positive correlation with Total_Credit_Cards and Total_Visits_Online and negative correlation with Total_calls_made.
- Total_Credit_Cards has strong positive correlation with Avg_Credit_Limit and negative correlation with Total_calls_made.
- Total_Visits_bank has strong negative correlation with Total_calls_made and Total_Visits_Online

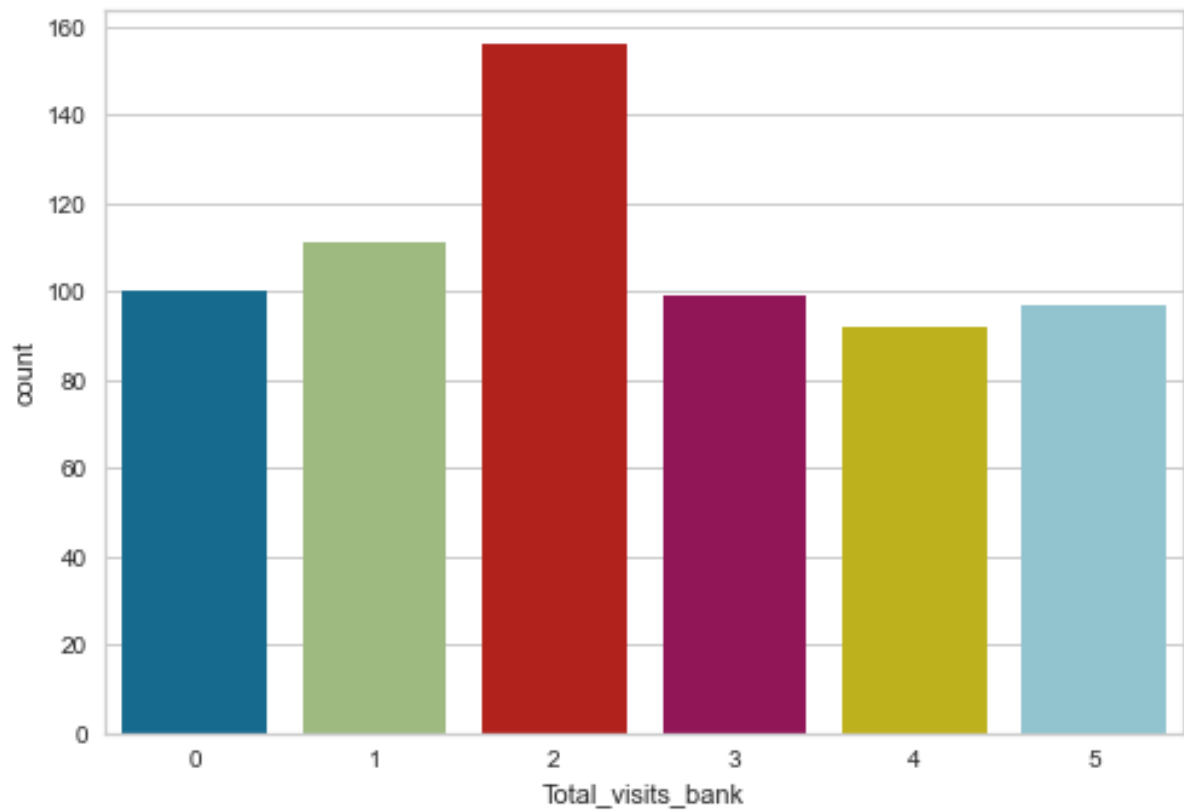


- We notice that Avg_Credit_Limit is showing a linear relationship with Total_Credit_Cards and Total_visits_online.
- From the pair plot, we see that there are atleast 2 peaks for some of the variables, which indicates that there would be a minimum of 2 clusters.



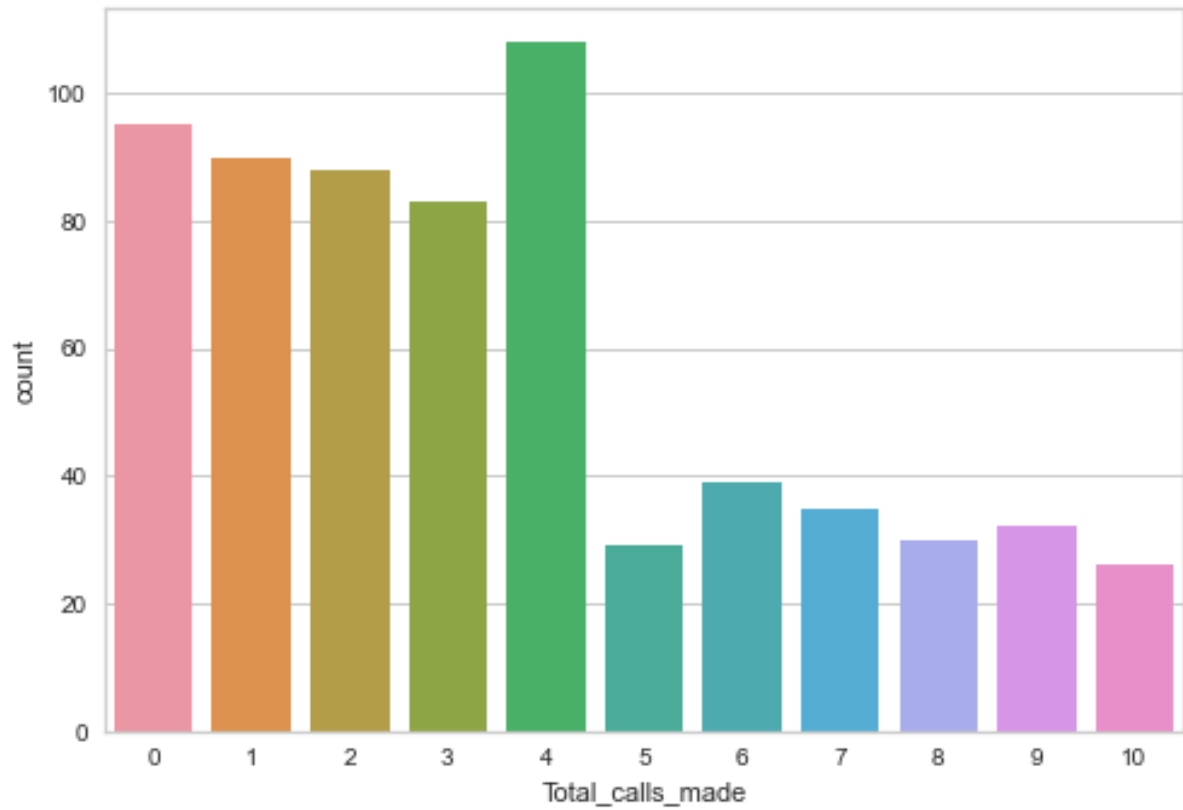
Observations:

- There are more customers(28.8%) who made 2 visits online for queries.
- There are atleast 20% of the customers who donot prefer to make online enqueries.



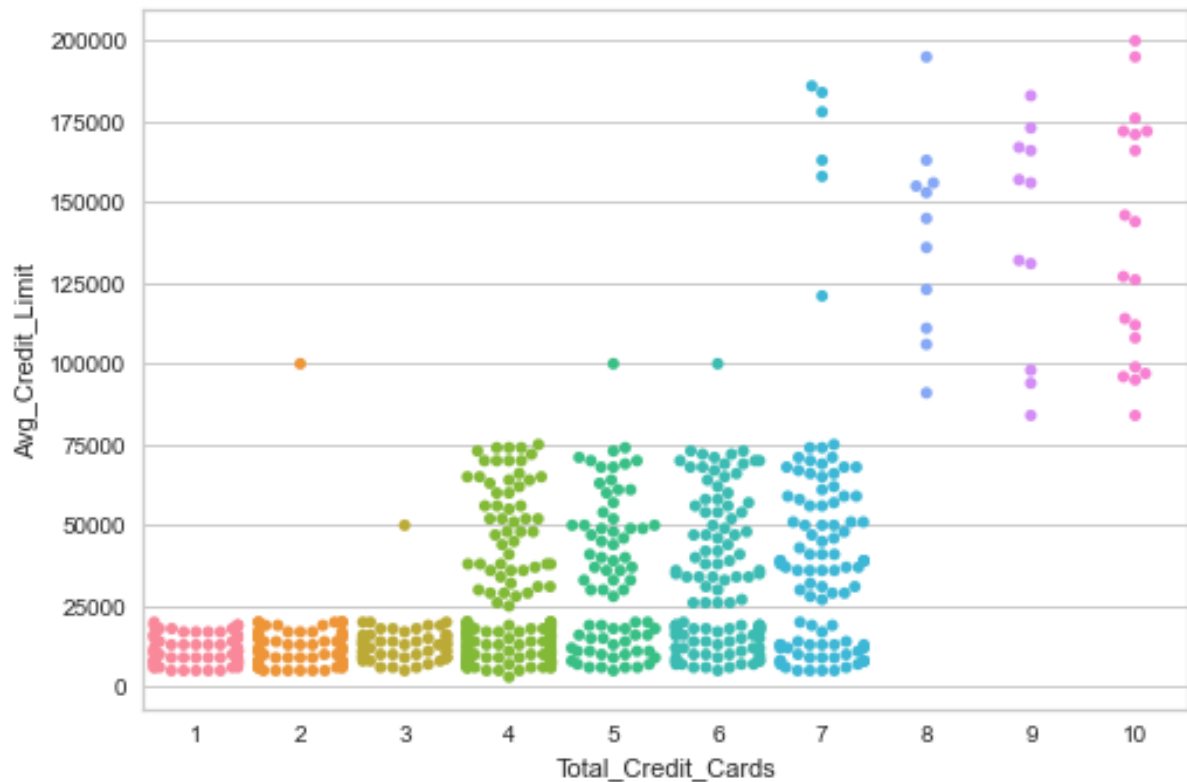
Observations:

- There are more customers(24%) who visit the bank twice.
- There are atleast 15% of the customers who donot prefer to visit the bank for enquiries.



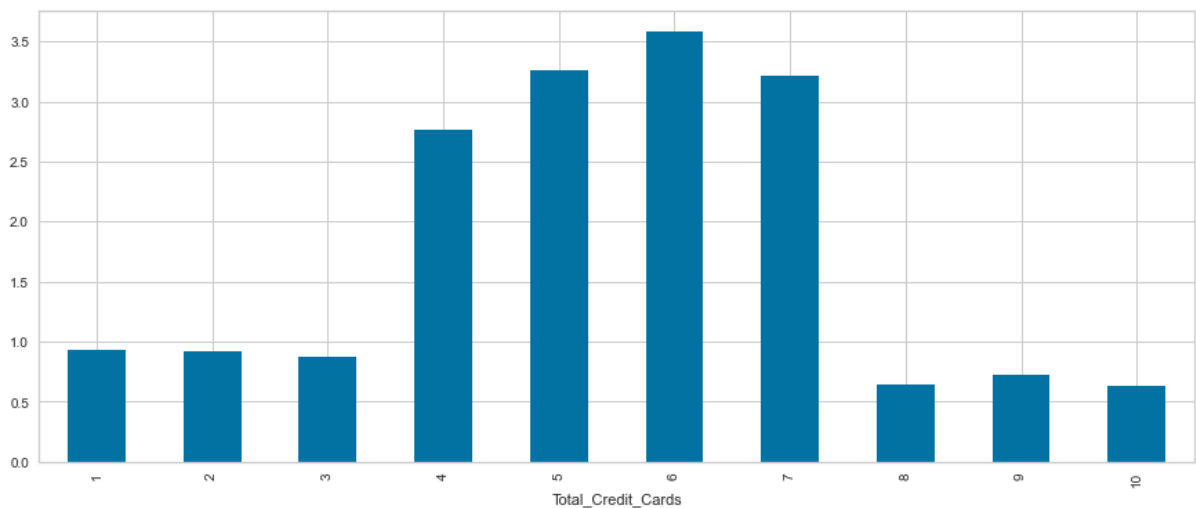
Observations:

- There are more customers(16%) who made 4 calls to the bank for queries.
- There are atleast 14.8% of the customers who donot prefer to make calls to the bank for enquiries.



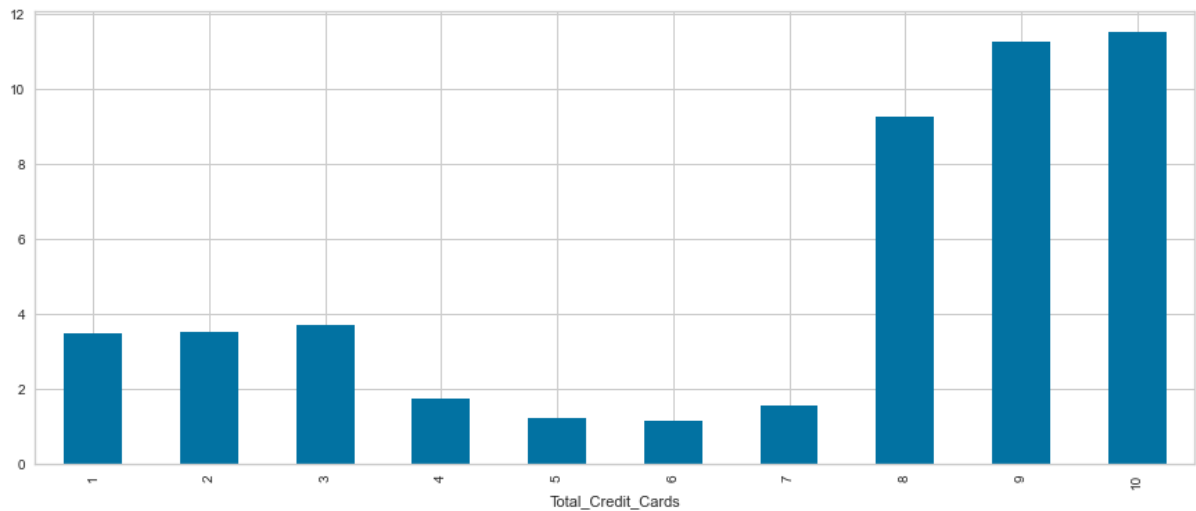
Observations:

- We notice a positive correlation between Total_Credit_Cards and Avg_Credit_Limit.
- Greater the number of cards, the higher the credit limit.



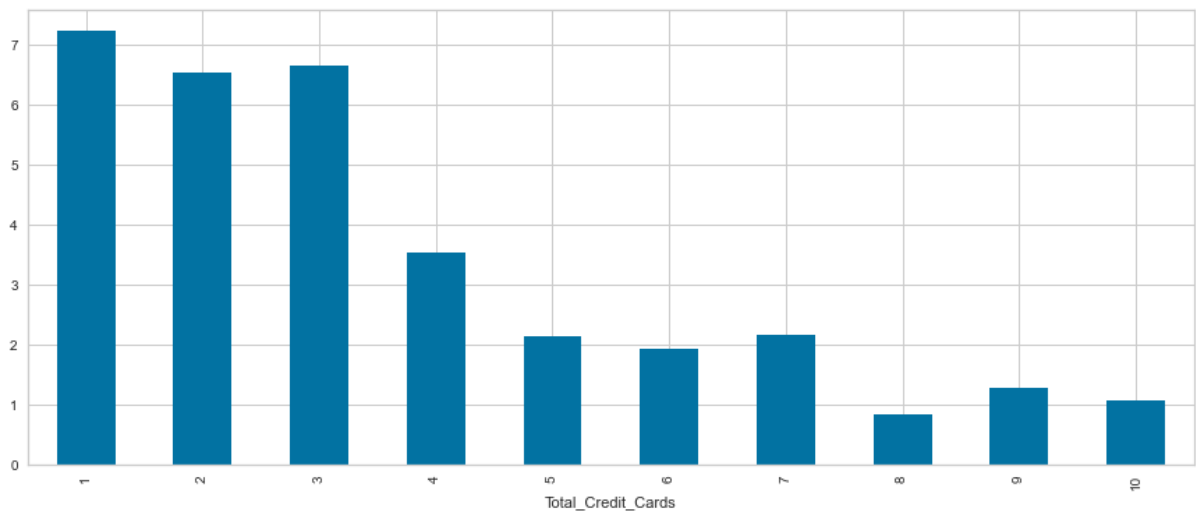
Observations:

- Customers with 4-7 cards have made more visits to the bank, followed by the customers who owned 1 – 3 cards.
- Customers with 8-10 cards made the least number of personal visits to the bank.



Observations:

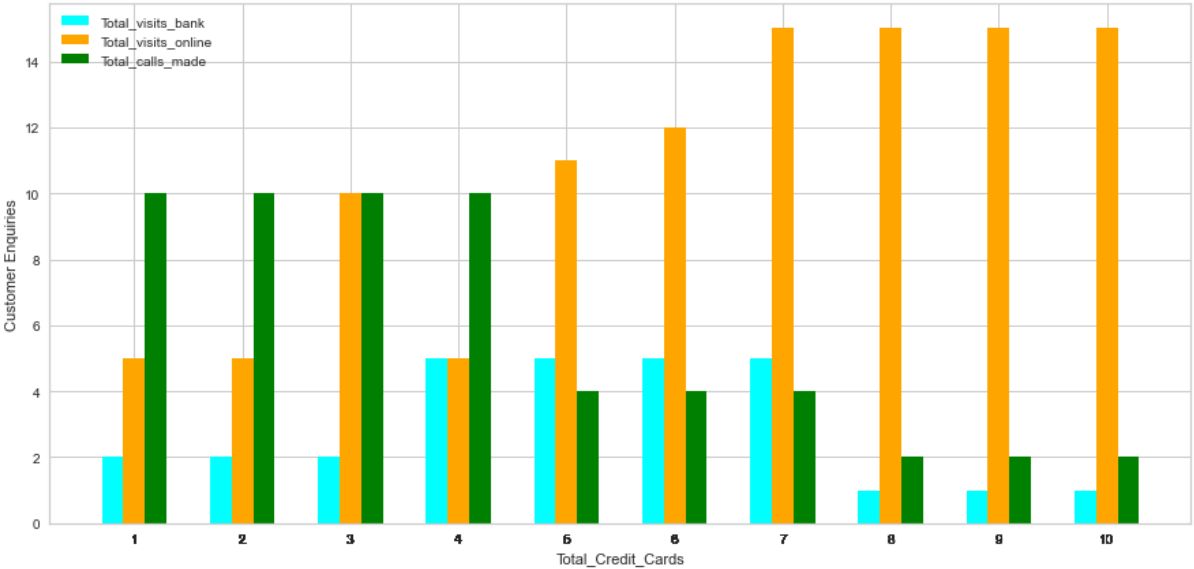
- Customers with 8-10 cards have made more online visits to the bank for enquiries.
- Customers with 1-3 cards made on an average 3 online visits to the bank for enquiries.
- Customers with 4-7 cards have made least number of online visits(one on an average) to the bank.



Observations:

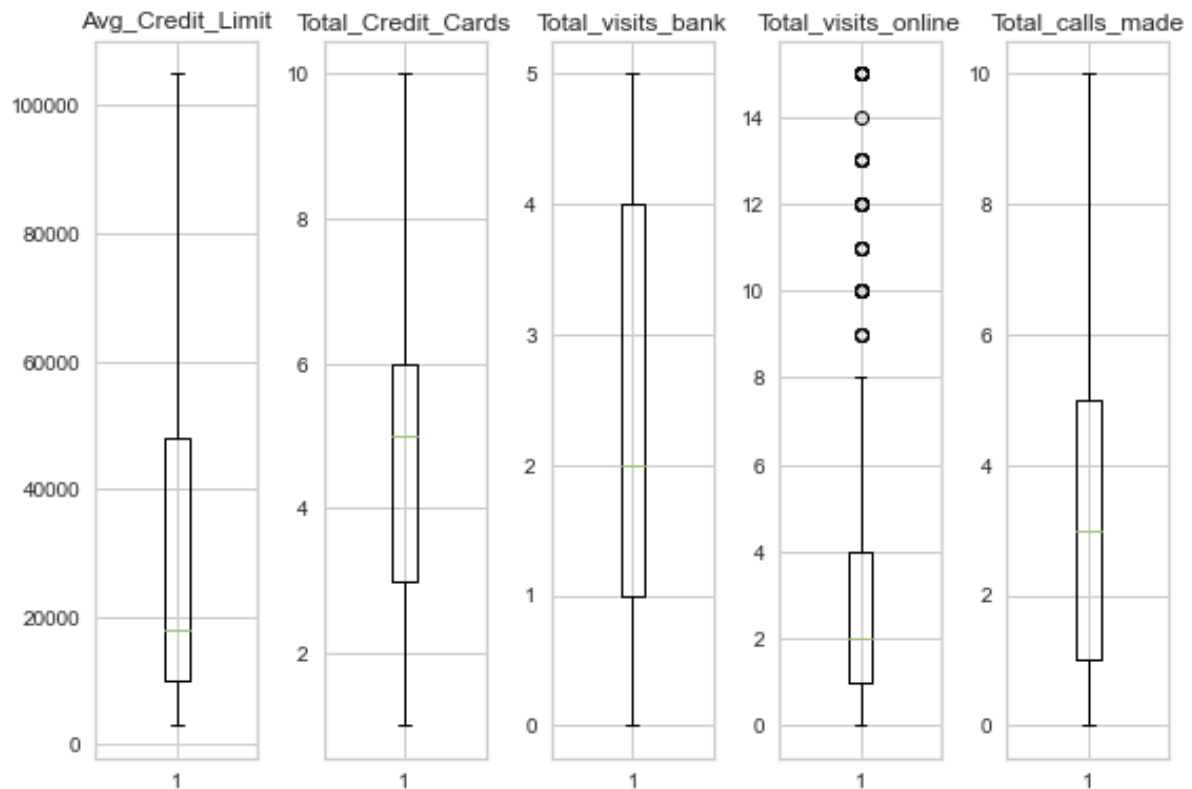
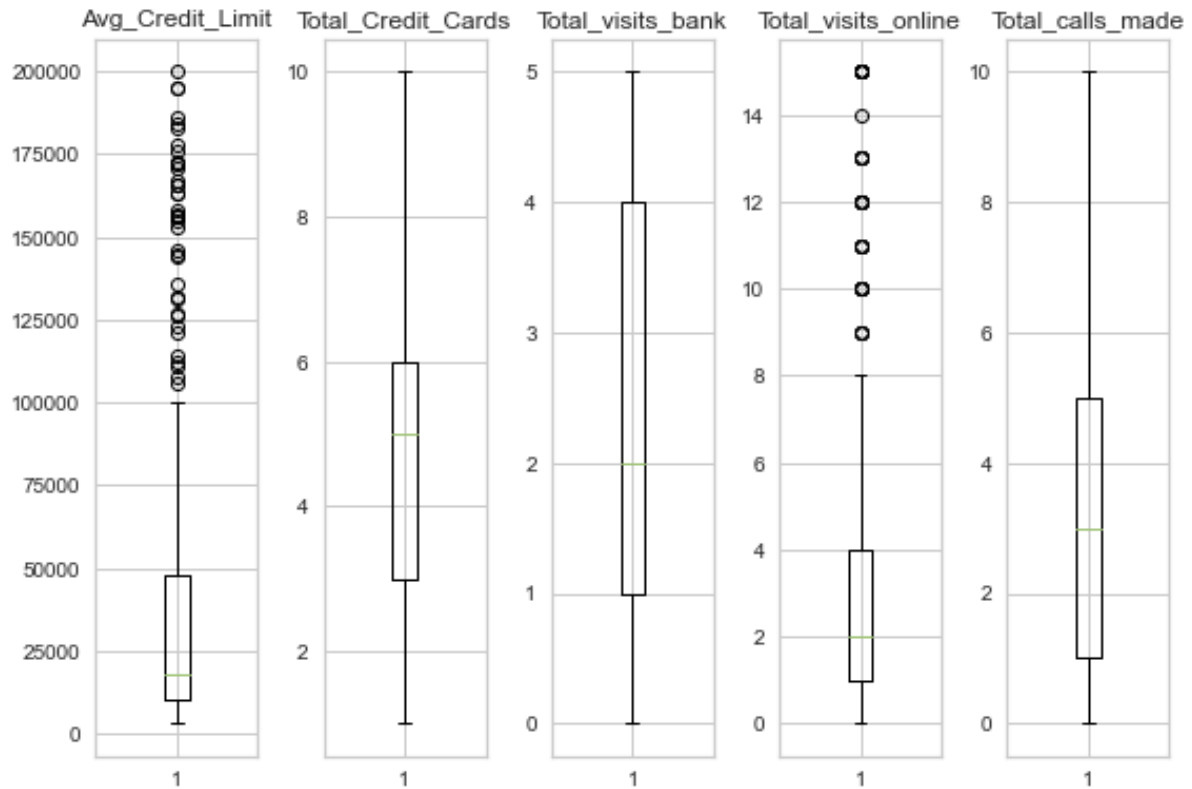
- Customers with 1-3 cards have made more calls compared to others.

All Life Bank Business Report



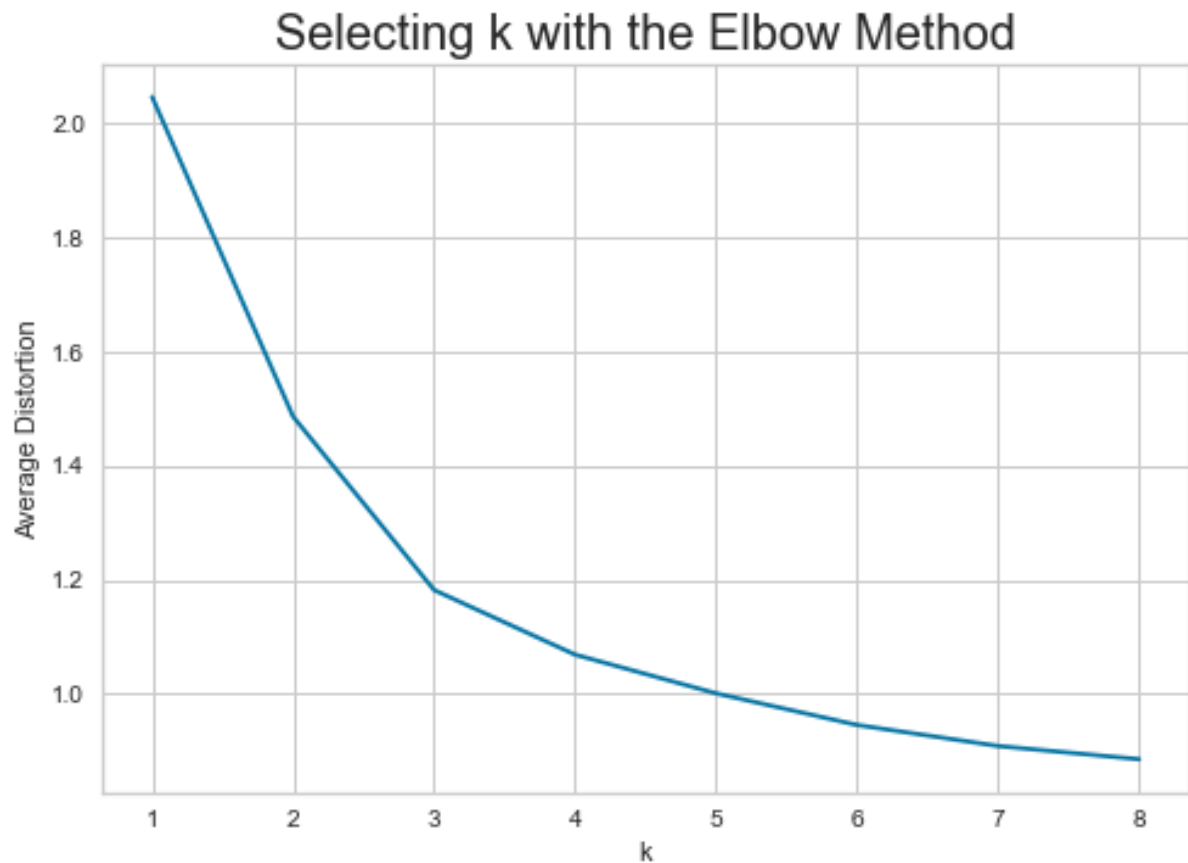
Treating Outliers

Lets look at outliers in every numerical column



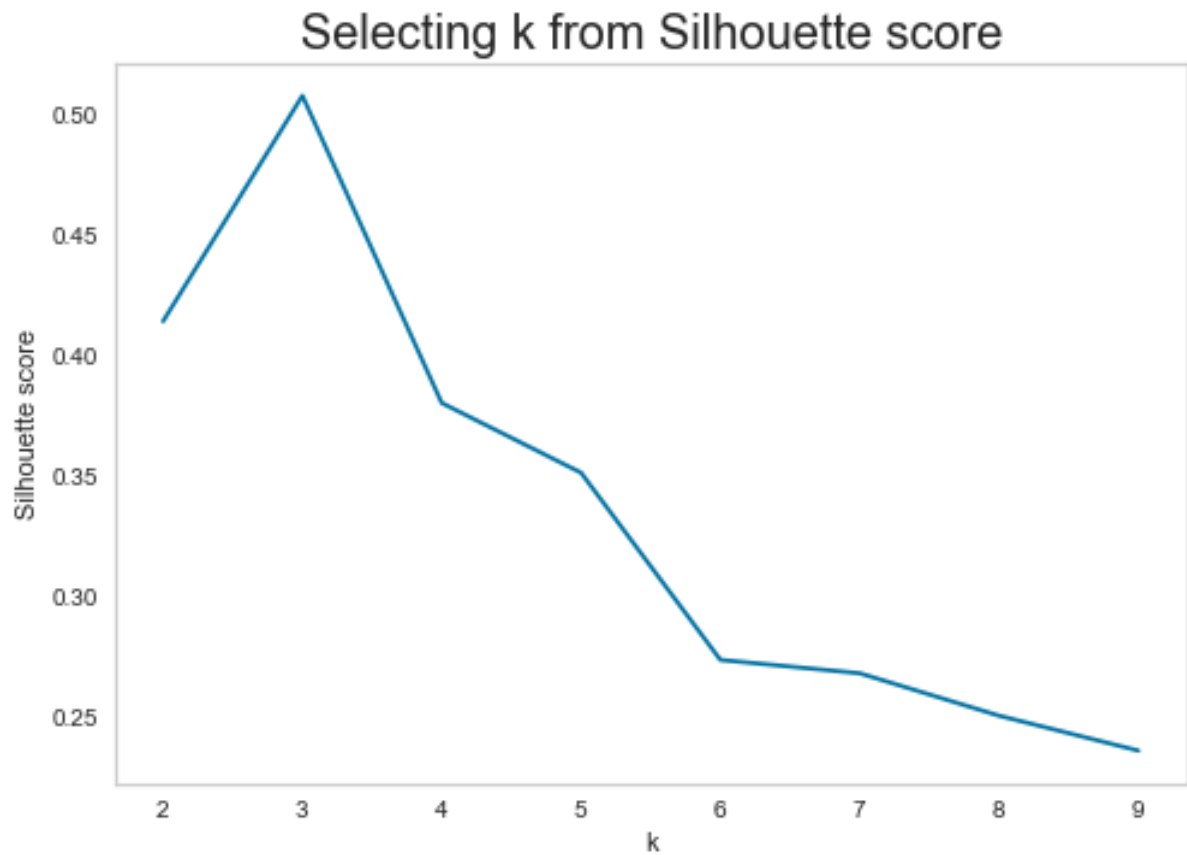
- We capped the values at 1.5IQR for only Avg_Credit_Limit column as it is a continuous variable.
- The outliers in Total_visits_online variable are not treated as its values are discrete in nature.

K-Means Clustering

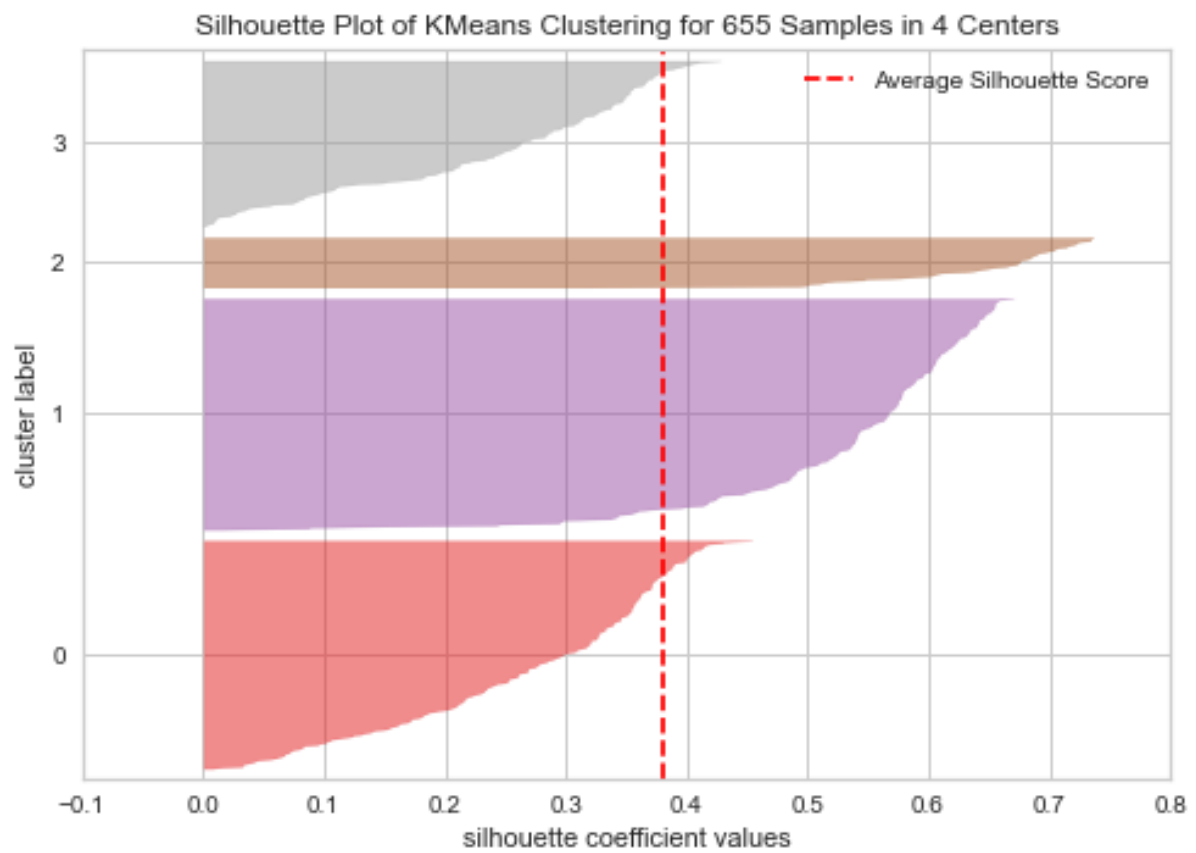
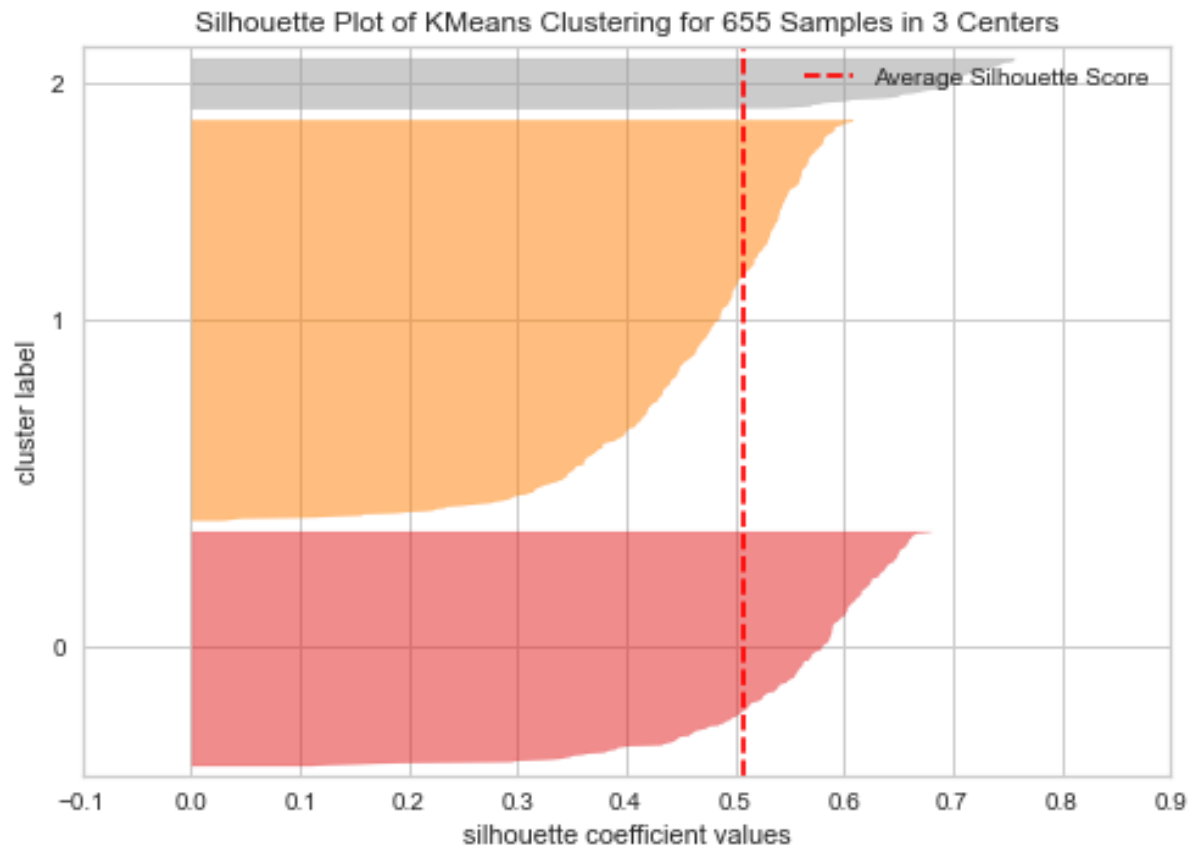


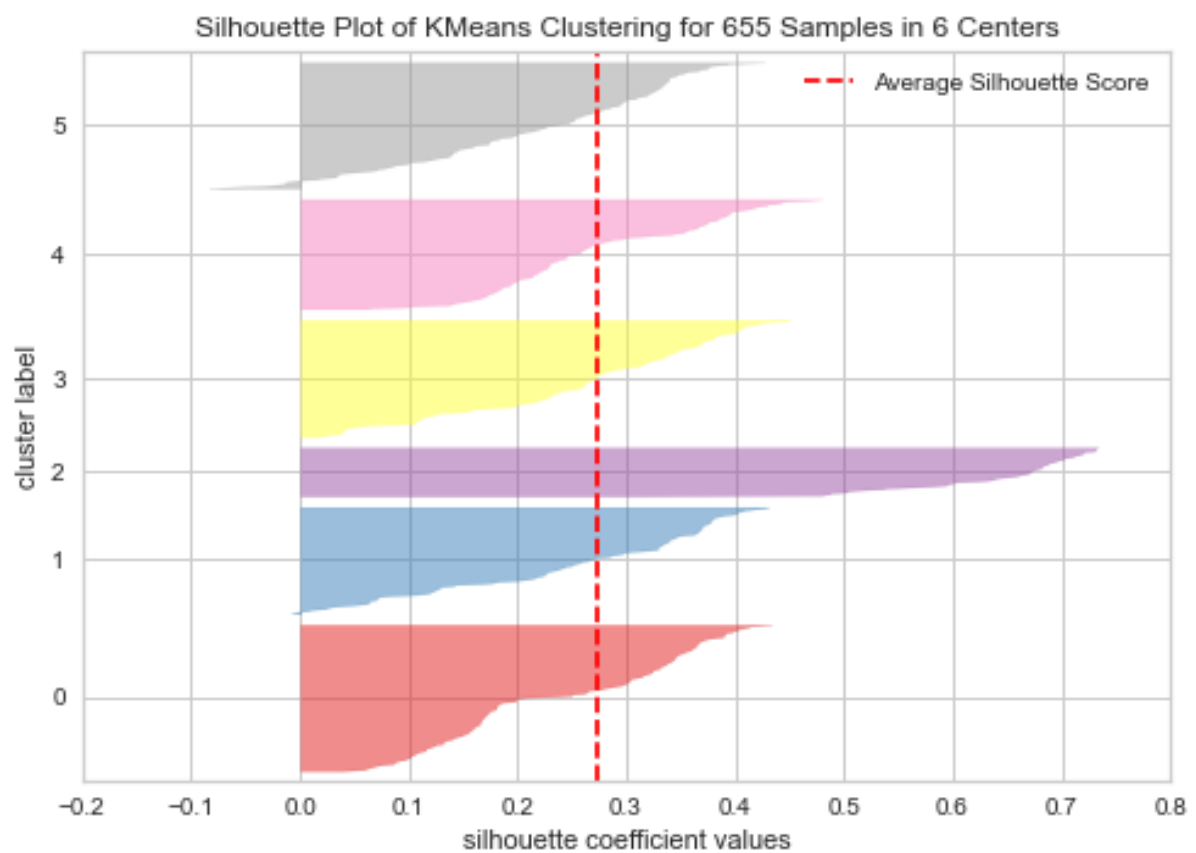
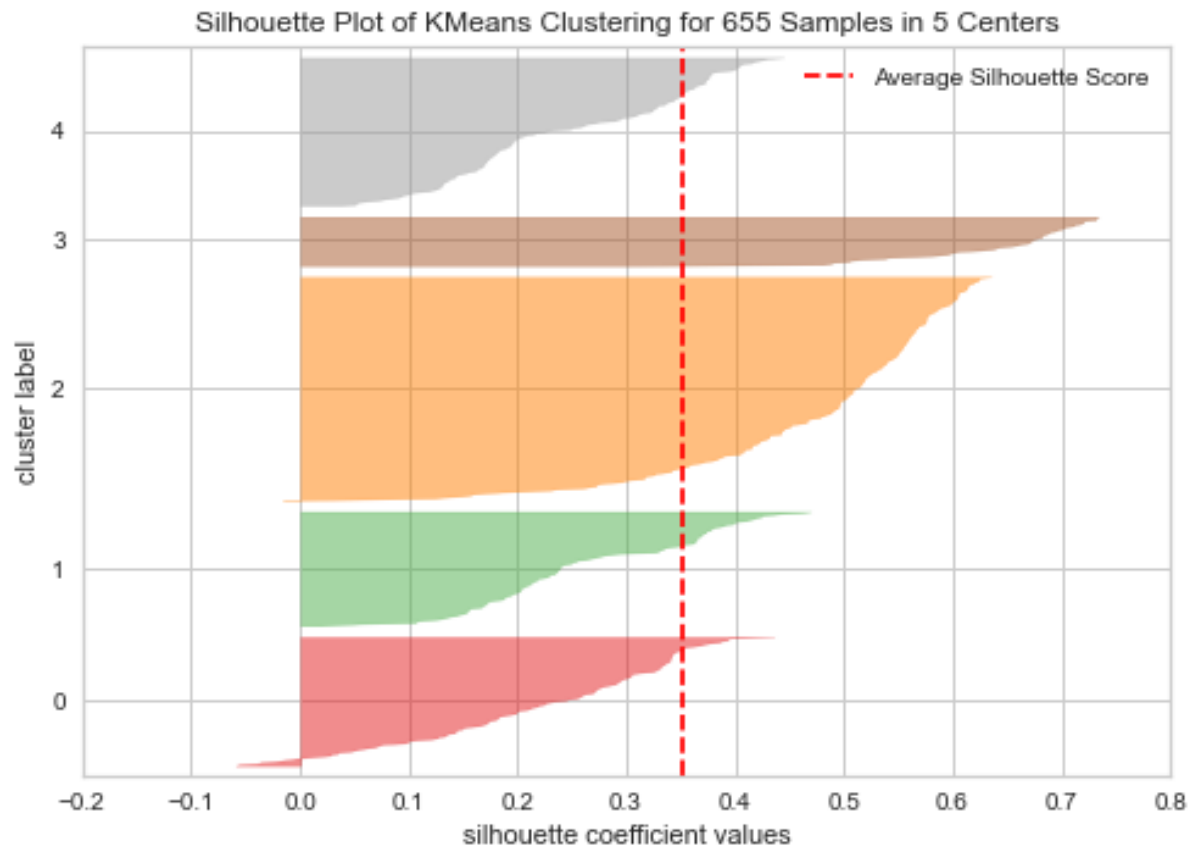
From Elbow method, 3 can be a good value for K

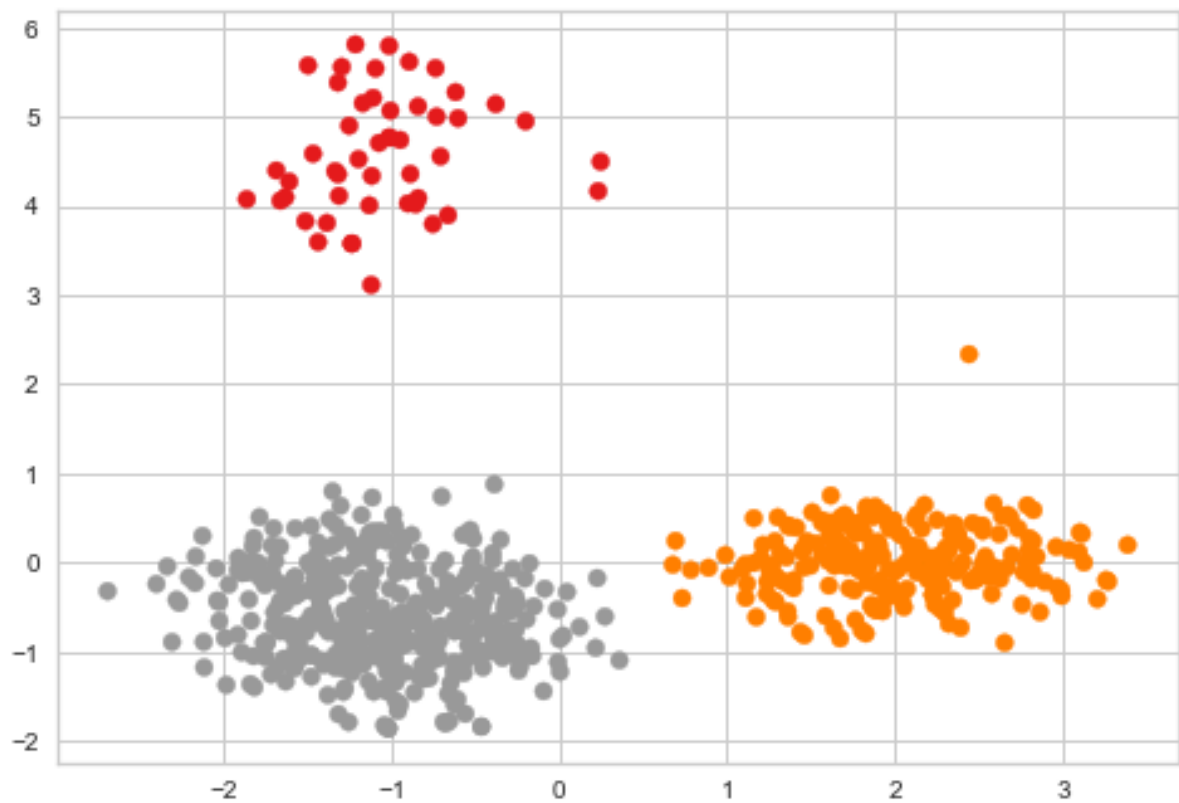
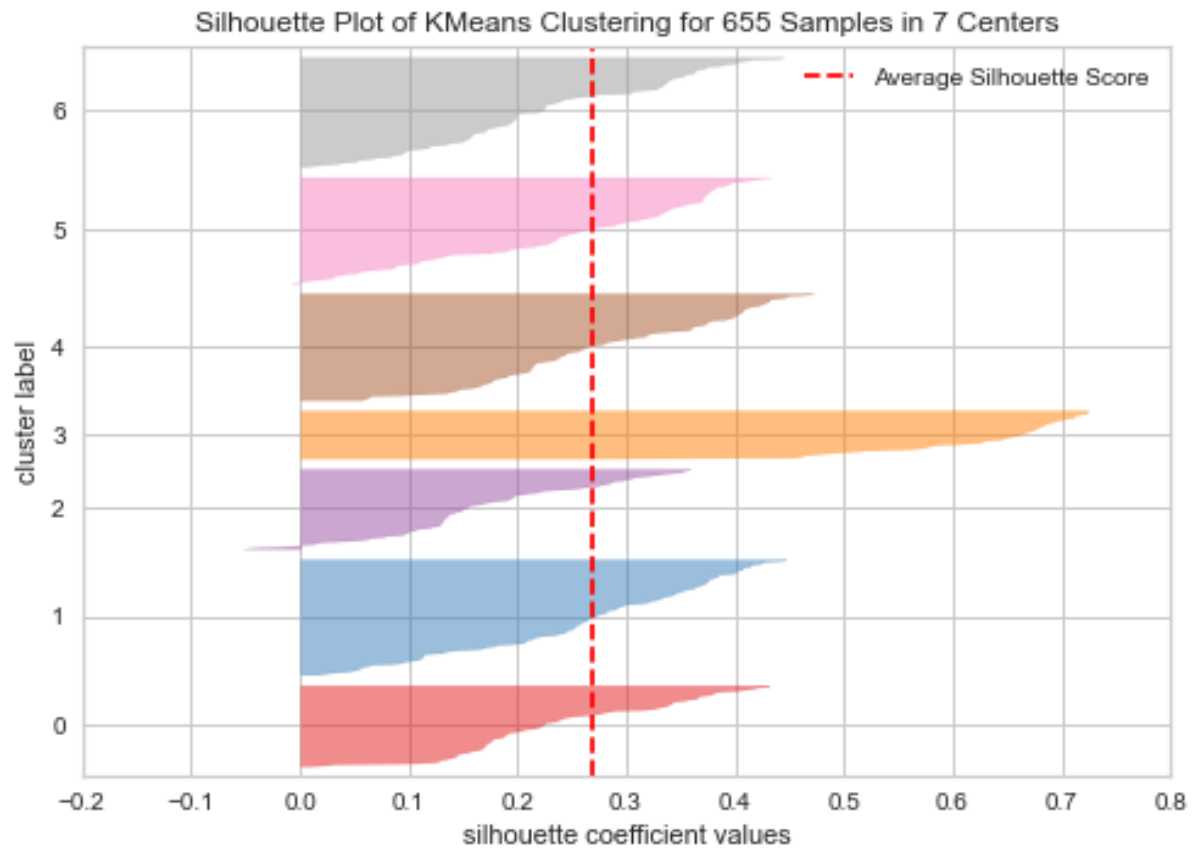
Lets us check silhoutte score



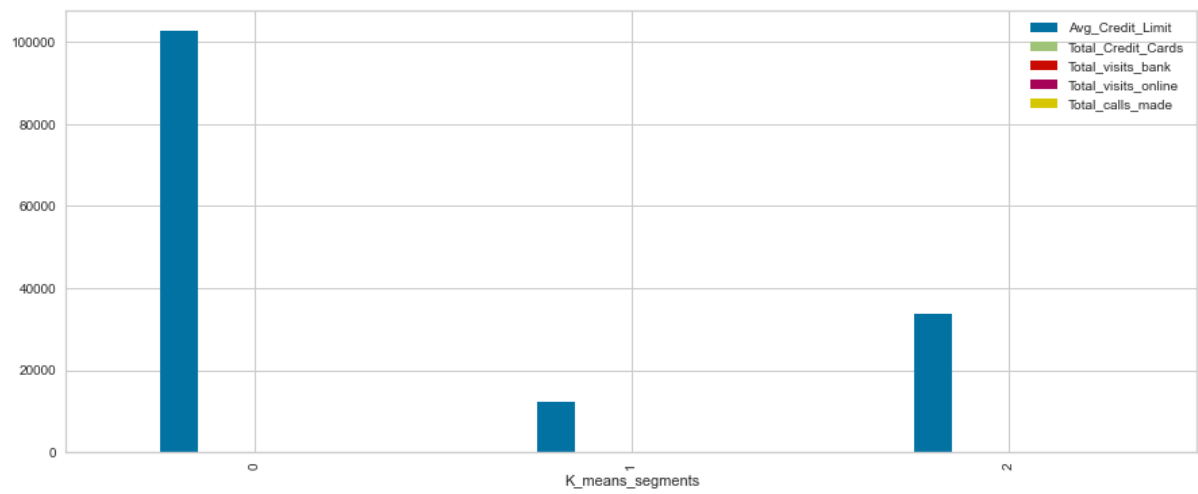
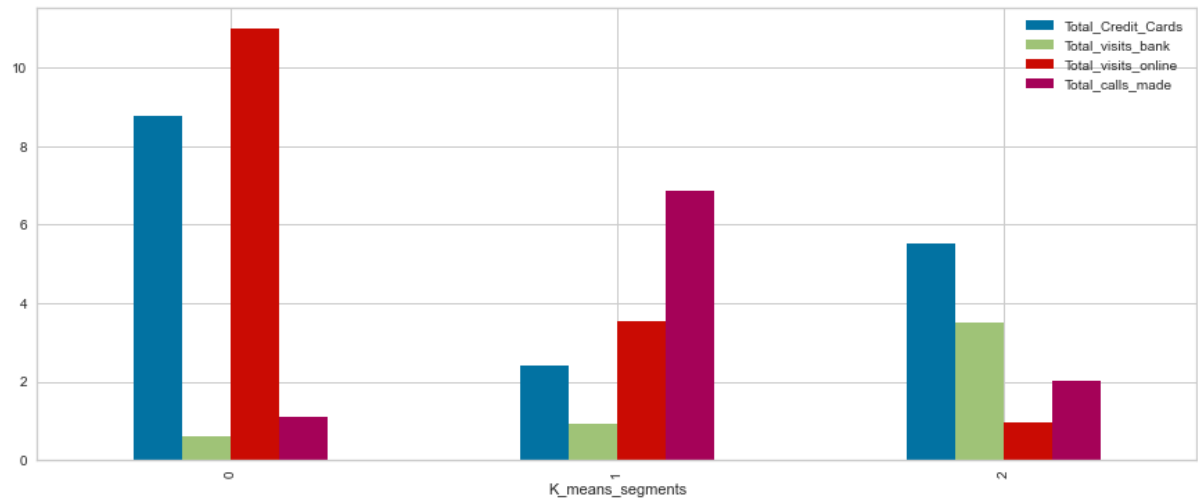
From silhouette score, it seems that 3 is good value of k



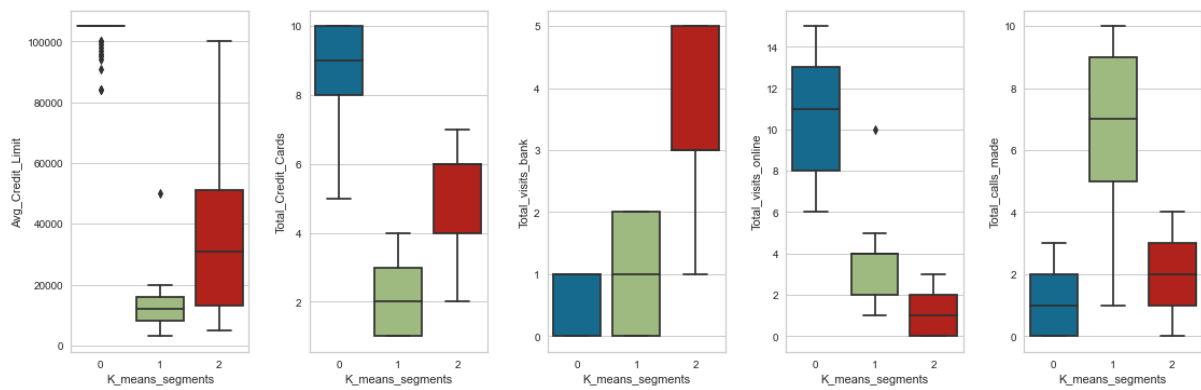




All Life Bank Business Report



Boxplot of numerical variables for each cluster



Insights

Cluster 0 (Platinum Customers) :

- Avg Credit limit is high for this cluster.
- Total credit cards taken by customers on an average very high.
- Total Visits to the bank is very very low.
- Total Visits online is highest, on an average 11 visits.
- Total calls made is lowest.

Cluster 1 (Silver Customers) :

- Avg Credit limit is low for this cluster.
- Total credit cards taken by customers is very low.
- Total Visits to the bank is low.
- Total Visits online is medium, on an average 3 visits.
- Total calls made is very high compared to other clusters.

Clusters 2 (Gold Customers):

- Avg Credit limit is medium for this cluster
- Total credit cards taken by customers on an average is medium.
- Total Visits to the bank is very high.
- Total Visits online is lowest as minimum as 1.
- Total calls made is lower.

Business Recommendations:

Platinum Customers-

Avg Credit limit and Total credit cards taken by customers on an average is very high. These customers prefer online visits over phone call and bank visits.

Recommendation –

- These customers already have more than average number of cards. AllLife bank should not focus on upselling to these customers, but rather focus on improving the customer service to retain them.
- These customers prefer online banking, so AllLife should focus on improving the user userface to improve customer experience. It is also recommended that the bank uses surveys to get feedback from these customers and continually improve the FAQs section to answer all the questions.
- It is also recommended to provide 24x7 online customer support in the form of live chat and emails. AllLife bank can also distinguish itself from other banks by providing a guaranteed response within a short window.
- The platinum customers own a large number of credit cards and can be great ambassadors to the bank to attract newer customers through referrals. AllLife bank should provide incentives to these customers for providing referrals.
- Investing on an app is also a good idea to attract new customers who prefer mobile banking.

Silver Customers-

Avg Credit limit and Total credit cards taken by customers on an average is very low. These customers prefer phone calls over Online and bank visits.

Recommendations-

- These customers have the lowest number of credit cards; there is an opportunity to sell them more by improving their experience with the bank. AllLife bank should, therefore, train the customer service representatives and empower them to resolve the customer issues to reduce the number of phone calls and the time needed to resolve the issues.
- In addition to improving the customer service, AllLife bank can provide additional perks to customers who own greater number of credit cards to encourage existing customers to apply for more credit cards.
- AllLife bank should hire multilingual representatives to cater to customers who speak different languages.

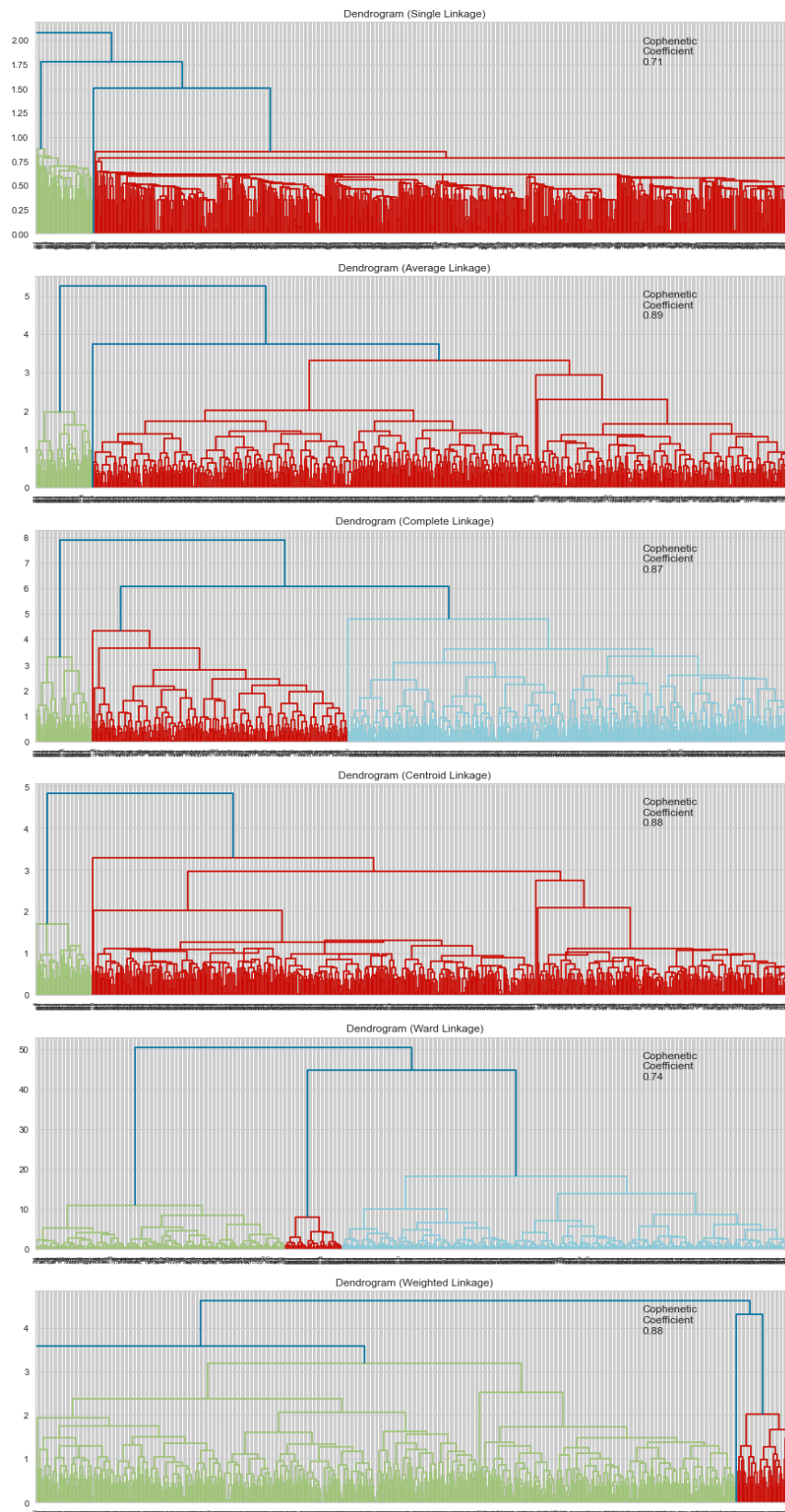
Gold Customers-

Avg Credit limit and Total credit cards taken by customers on an average is medium. These customers prefer bank visits over phone calls and online.

Recommendations-

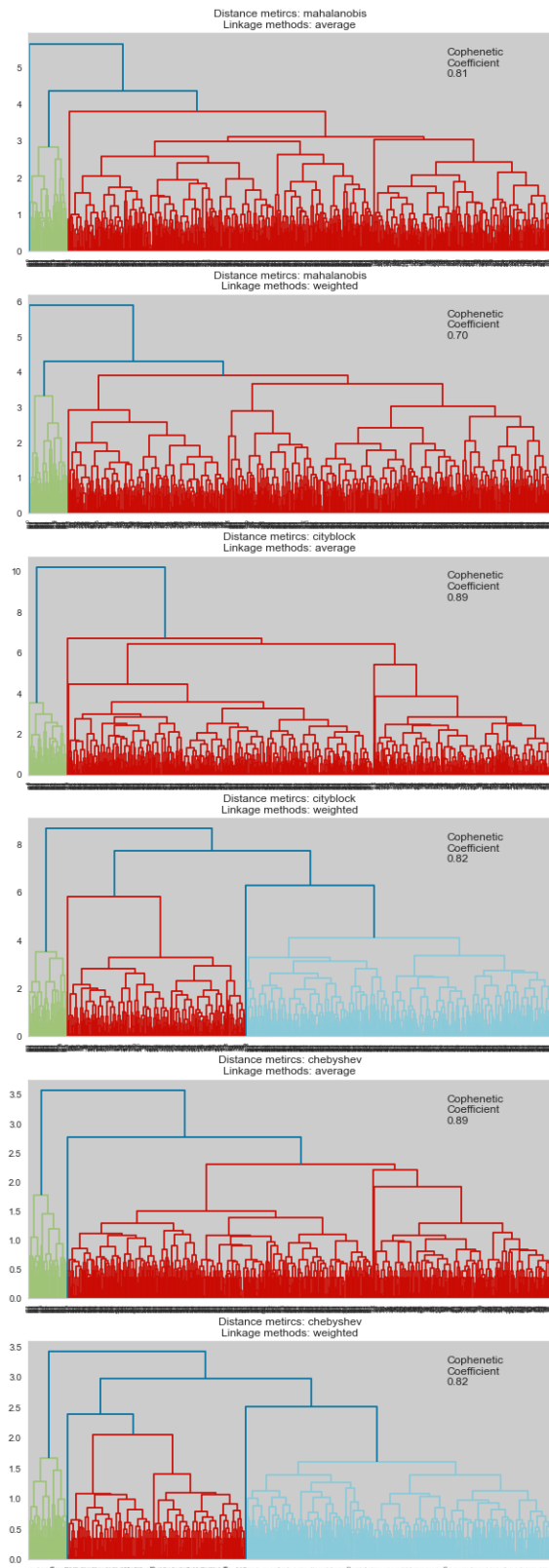
- The Gold Customers prefer personal visits to the bank. AllLife bank can improve the customer experience by Encouraging customers to schedule appointments to reduce wait times.
- Building a luxury lounge in the bank with amenities (TVs, coffee, internet) for walk in (no appointment) customers while they wait for service.
- Following up with the customers after their visit to check if their issues have been resolved. This provides AllLife representatives an opportunity to address any unresolved issues and improve customer experience and loyalty.

Hierarchical Clustering



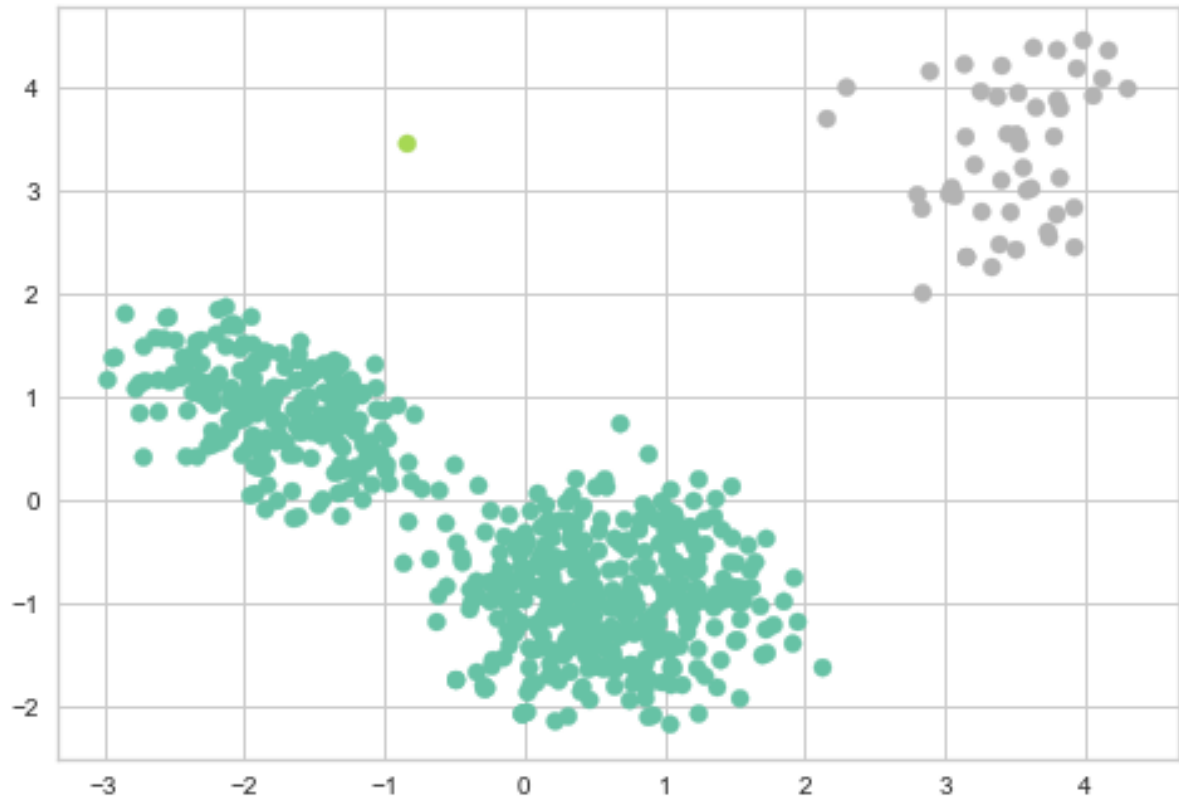
Observations:

- Dendrogram with single and ward linkage method are showing distinct and separate clusters.



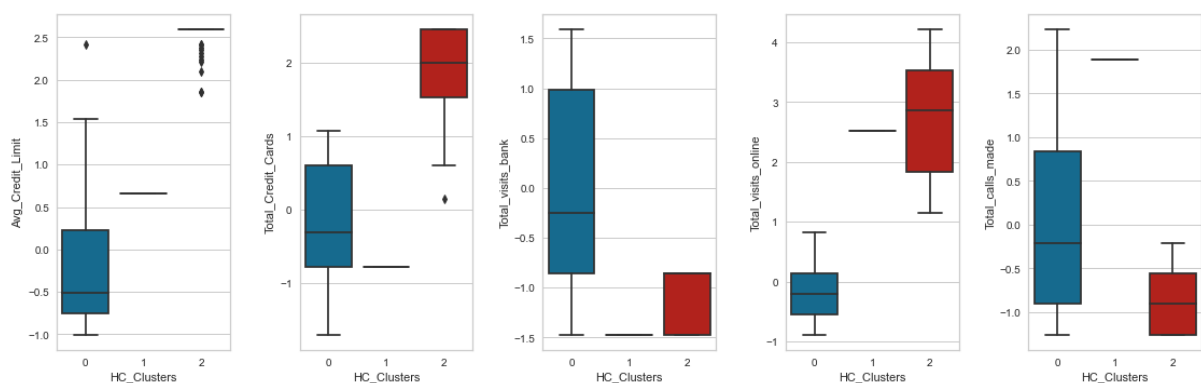
Observations:

- Out of all the dendrogram we saw, it is clear that dendrogram with weighted linkage method gave us separate and distinct clusters
- 3 cluster would be appropriate number of cluster from dendrogram with weighted linkage method



We notice that there is only one record in one cluster with single linkage method. Lets try Ward Linkage method.

Boxplot of numerical variables for each cluster



Insights from Single Linkage Method:

Cluster 0 (Silver Customers):

- Avg Credit limit is less for this cluster
- Total credit cards taken by customers on an average is medium.
- Total Visits to the bank is high.
- Total Visits online is lowest as minimum as 2.
- Total calls made is medium.

Cluster 1 (Gold Customers):

- Avg Credit limit is medium for this cluster.
- Total credit cards taken by customers on an average is less.
- Total Visits to the bank is zero.
- Total Visits online is high, on an average 10 visits.
- Total calls made is high.

Cluster 2 (Platinum Customers):

- Avg Credit limit is high for this cluster.
- Total credit cards taken by customers on an average very high.
- Total Visits to the bank is very low.
- Total Visits online is highest, on an average 11 visits.
- Total calls made is lowest.

Business Recommendations:

Platinum Customers -

Avg Credit limit and Total credit cards taken by customers on an average is very high. These customers prefer online visits over phone call and bank visits.

Recommendation –

- These customers already have more than average number of cards. AllLife bank should not focus on upselling to these customers, but rather focus on improving the customer service to retain them.

Silver Customers -

Avg Credit limit of customers on an average is very low and total number of credit cards taken is medium. These customers prefer bank visits over Online and phone calls.

Recommendations-

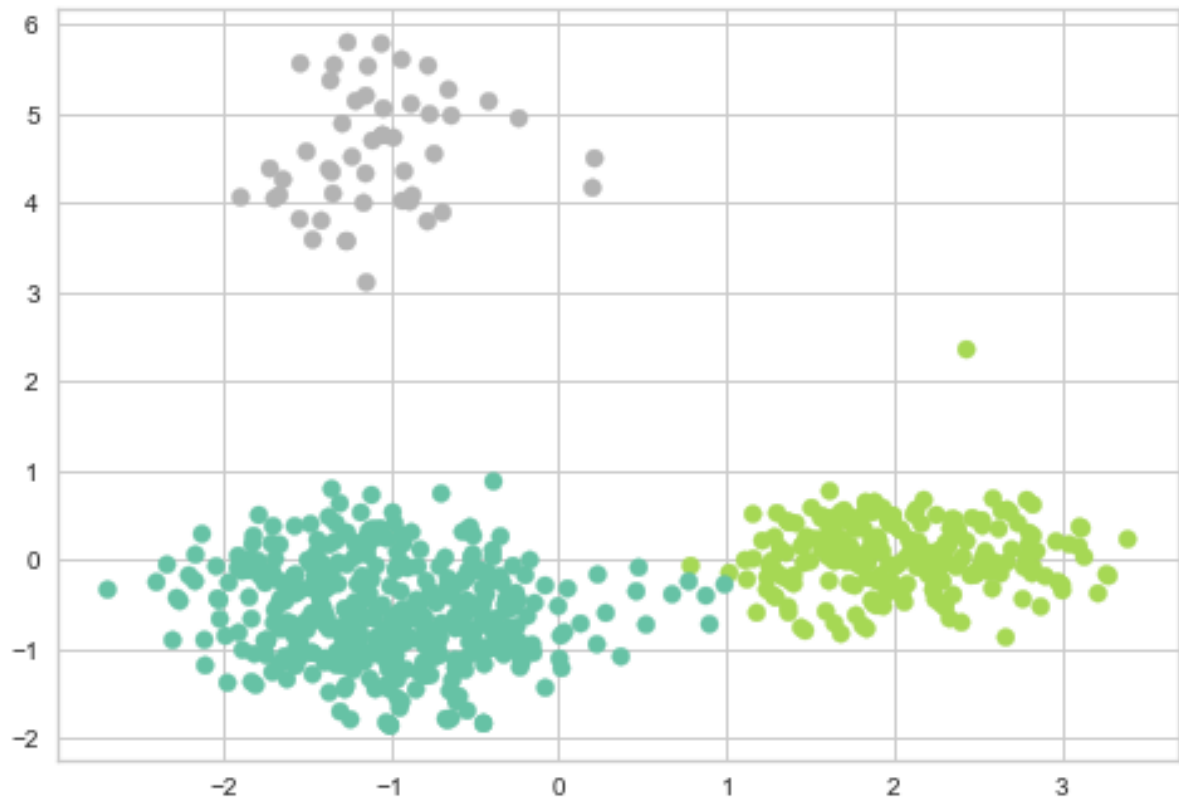
- These Customers prefer personal visits to the bank. AllLife bank can improve the customer experience by Encouraging customers to schedule appointments to reduce wait times.
- In addition to improving the customer service, AllLife bank can provide additional perks to customers who own greater number of credit cards to encourage existing customers to apply for more credit cards.

Gold Customers -

Avg Credit limit of customers on an average is very medium and total number of credit cards taken is less. These customers prefer Online banking and phone calls over bank visits.

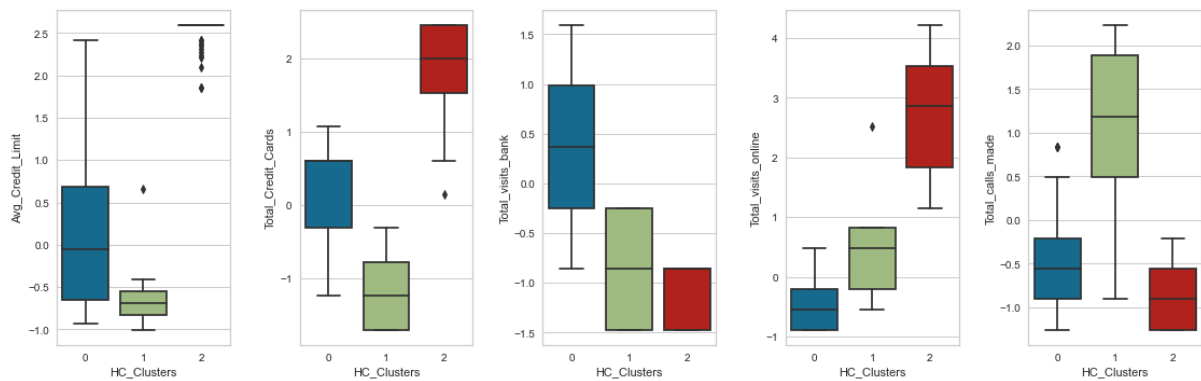
Recommendations-

- These customers have the lowest number of credit cards; there is an opportunity to sell them more by improving their experience with the bank.
- Investing on an app is also a good idea to attract new customers who prefer mobile banking.
- AllLife bank should hire multilingual representatives to cater to customers who speak different languages.



We notice that 3 distinct clusters are formed with no overlapping and the results are almost similar to K-Means.

Boxplot of numerical variables for each cluster



Insights from Ward Linkage Method:

Cluster 0 (Gold Customers):

- Avg Credit limit is medium for this cluster
- Total credit cards taken by customers on an average is medium.
- Total Visits to the bank is high.
- Total Visits online is lowest as minimum as 1.
- Total calls made is medium.

Cluster 1 (Silver Customers):

- Avg Credit limit is low for this cluster.
- Total credit cards taken by customers is less.
- Total Visits to the bank is medium.
- Total Visits online is medium, on an average 3 visits.
- Total calls made is high.

Cluster 2 (Platinum Customers):

- Avg Credit limit is high for this cluster.
- Total credit cards taken by customers on an average very high.
- Total Visits to the bank is very low.
- Total Visits online is highest, on an average 11 visits.
- Total calls made is lowest.

Observations:

- There are more customers in cluster 0 with 4-7 Credit cards
- There are more customers in cluster 1 with 1-3 creditcards
- There are more customers in cluster 2 with 8-10 creditcards.

Cluster Comparisons

- K-Means and Hierarchical Clustering- Ward Linkage Methods shows similar results.
- However, Hierarchical Clustering – Single Linkage method gives different set of clusters with different averages and different grouping. The variance in one group is extremely high and there is no uniform distribution and there seems to be an overlap in total visits online from cluster 1 and 2.
- In “Platinum Customer” cluster, 49 customers are identified as platinum customers in all the methods.
- In “Gold Customer” cluster, 382 potential members are identified through K- Means clustering method and 390 members through Hierarchical clustering-Ward Linkage method and 605 customers in Hierarchical clustering- Single Linkage method .
- In “Silver Customer” cluster, 224 members are identified through K- Means clustering method and 216 members through Hierarchical clustering method and 1 customer in Hierarchical clustering-Single Linkage method .
- The differences in Gold and Silver Clusters from both the K-Means and Hierarchical Clustering- Ward Linkage Methods are very close.
- As we know, k-means algorithms assume that clusters are spherical in shapes but that is not the assumption of HC algorithm.
- K-Means clustering gave us a fast solution on k value without much workaround, though we had to initialize and try on multiple k values.
- With K-Means clustering, both Elbow method and Silhouette score confirmed that k value=3 is apt.
- However, there was little workaround with Hierarchical clustering to conclude on number of clusters as the single linkage method lead us with one of the clusters with a single record which doesn't seem to be appropriate as there is an overlap in some values and thereby, we had to choose ward linkage method. With ward linkage method, we had to try both 4 clusters and 3 clusters and then conclude on 3 clusters as there was an overlap between values with 4 clusters.

Therefore K-Means Clustering method worked well for this dataset.

PCA

Bartlett's Test of Sphericity

Checks for reasonable correlation between the observed variables. PCA is applied if there is a significant correlation.

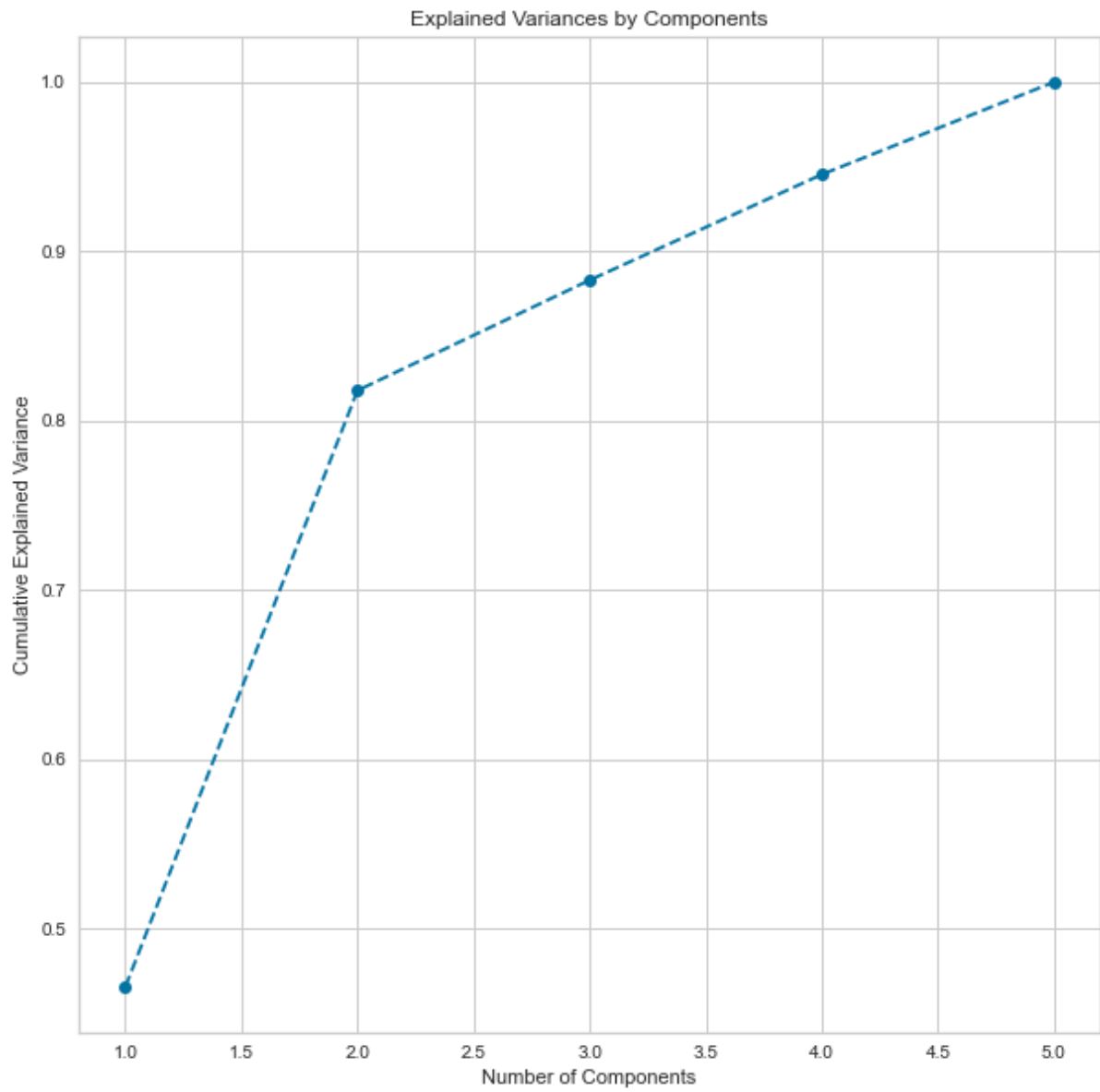
If Null Hypothesis is rejected it means there is a significant correlation and PCA can be applied.

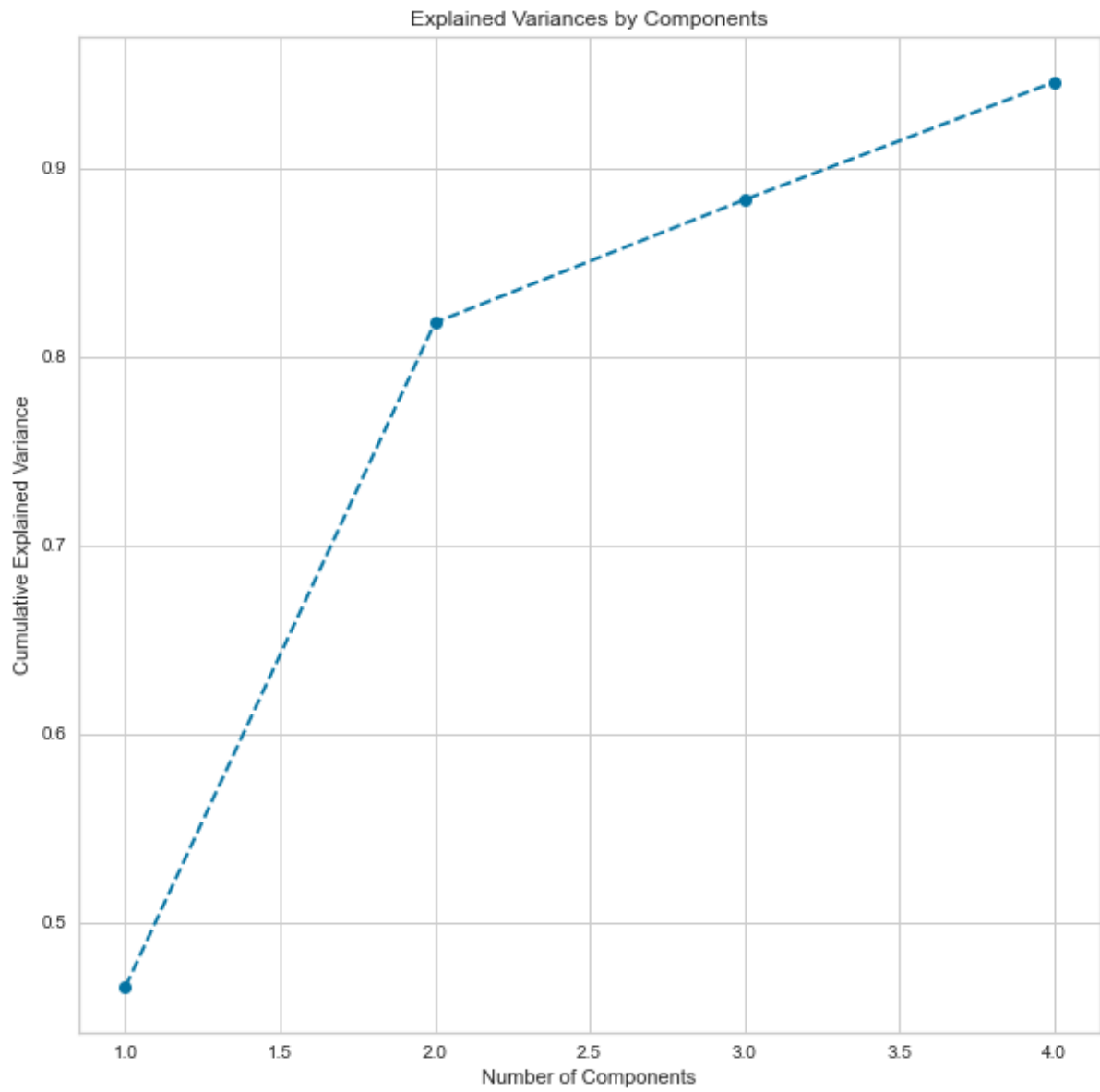
Hence we expect $p\text{-value} < 0.05$ from the test to proceed further

KMO Test

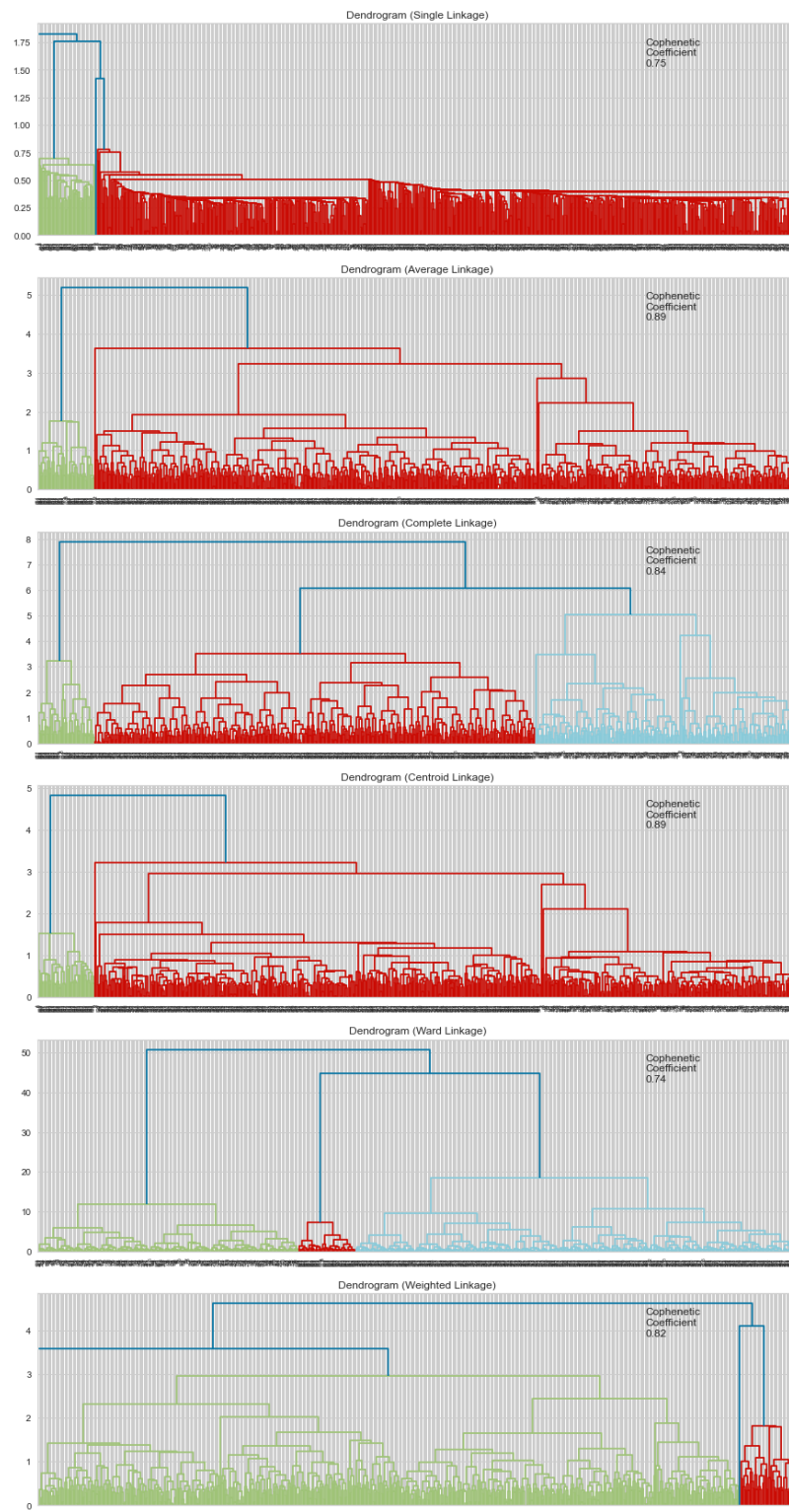
It used to check for sample adequacy before proceeding with PCA

KMO of greater than 0.5 indicates that sample size is adequate for getting the components





Perform Clustering



Business Recommendations:

Platinum Customers-

Avg Credit limit and Total credit cards taken by customers on an average is very high. These customers prefer online visits over phone call and bank visits.

Recommendation –

- These customers already have more than average number of cards. AllLife bank should not focus on upselling to these customers, but rather focus on improving the customer service to retain them.
- These customers prefer online banking, so AllLife should focus on improving the user interface to improve customer experience. It is also recommended that the bank uses surveys to get feedback from these customers and continually improve the FAQs section to answer all the questions.
- It is also recommended to provide 24x7 online customer support in the form of live chat and emails. AllLife bank can also distinguish itself from other banks by providing a guaranteed response within a short window.
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Recommendations-

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THANK YOU